

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\DMW Preheater Stack.pvdb

Date of Analysis : Sep 7, 2026 5:06pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1090.000 mm. ID
 Vessel Design Length, Tangent to Tangent 6360.80 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 150 °C
 Channel Side Design Temperature 200 °C
 Shell Side Design Pressure 10.000 bars
 Channel Side Design Pressure 191.000 bars
 Shop Shell Side Test Pressure 17.750 bars
 Shop Channel Side Test Pressure 257.503 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 304	...	...	S30400	+IT
Head	SA-240 304	...	...	S30400	
Head	SA-240 304	...	...	S30400	+IT
Nozzle	SA-312 TP304	...	...	S30400	
Nozzle	SA-312 TP304	...	...	S30400	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-240 304	...	...	S30400	
Nozzle Flg	SA-182 F304	...	t <= 130	S30400	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-240 304	...	...	S30400	
Tubes	SA-213 TP304	...	...	S30400	
Tubesheet	SA-965 F304	...	...	S30400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-387 11	2	...	K11789	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch Dish Head	191.004	0.00	194.000	0.0000	N/A	No
Ch Shell	191.004	0.00	285.000	0.0000	N/A	No
Main Shell	10.107	0.00	20.000	0.0000	N/A	No
Shell Dish End	10.107	0.00	22.000	0.0000	No	No

Liquid Level: 1090.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Stiffener Ring Specifications:

Elevation mm.	Selected Type	Ring Description
2670.00	Bar 110.6 x 30.0	SR, 910.0 Fr30
5460.00	Bar 110.6 x 30.0	SR, 3800.0 Fr30

Element Types and Properties:

Element	"To" Elev	Element Length	Nominal Thickness	Finished Thickness	Reqd Thk Internal	Reqd Thk External	Long Eff	Circ Eff
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Type	mm.	mm.	mm.	mm.	mm.	mm.		
Sphere	0.0	0.0	52.0	46.0	45.3	...	1.00	1.00
Cylinder	1500.0	1500.0	135.0	135.0	87.0	...	1.00	1.00
Cylinder	6310.0	4550.0	8.0	8.0	4.0	...	1.00	1.00
Ellipse	6360.8	50.8	10.0	8.0	3.6	...	1.00	1.00

Local Stress Analysis Results:

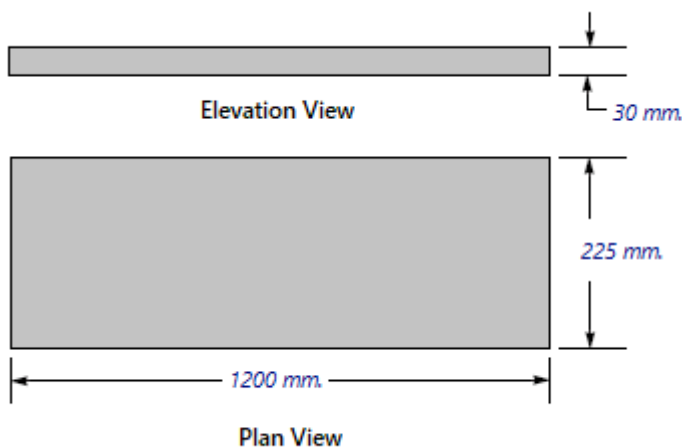
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A1-Tube Inlet	WRC-297	0.837	Nozzle	Passed
B1-Tube Outlet	WRC-297	0.837	Nozzle	Passed
A2-Shell inlet	WRC-297	0.493	Shell	Passed
B2-Shell outle	WRC-297	0.500	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm.
Saddle Bearing Angle	167.000 deg.
Centerline Dimension	1300.000 mm.
Wear Pad Width	350.000 mm.
Wear Pad Thickness	8.000 mm.
Wear Pad Bearing Angle	180.000 deg.
Distance from Saddle to Tangent	910.000 mm.
Baseplate Length	1200.000 mm.
Baseplate Thickness	30.000 mm.
Baseplate Width	225.000 mm.
Number of Ribs (including outside ribs)	4
Rib Thickness	20.000 mm.
Web Thickness	20.000 mm.
Height of Center Web	570.000 mm.
Number of Bolts in Baseplate	2

Baseplate Sketch



Maximum Tensile Bolt Load 1944. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	85221.27 Kgf
Maximum Transverse Saddle Shear Load	7818.91 Kgf
Maximum Longitudinal Saddle Shear Load	4975.45 Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	8085.14 Kgf
Maximum Transverse Saddle Shear Load	433.02 Kgf

Maximum Longitudinal Saddle Shear Load 81.28 Kg

Weights:

Fabricated - Bare w/o removable internals	43915.8	kg.
Shop Test - Fabricated + water (full)	55173.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	43915.8	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	43915.8	kg.
Empty - Fab. + intls. + details + wghts.	43915.8	kg.
Operating - Empty + operating liquid (no ca)	51717.1	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
M1-Man Hole	194	...	...	OK	OK	Passed
A1-Tube Inlet	280	...	...	OK	OK	Passed
B1-Tube Outlet	280	...	...	OK	OK	Passed
A2-Shell inlet	20	...	...	OK	OK	Passed
B2-Shell outlet	20	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 20 Nozzle : B2-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 194 Nozzle : M1-Man Hole [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Inside Radius mm.
10	M1-Man Hole	0.000	0.000	88.525
20	A1-Tube Inlet	750.000	90.000	130.600
20	B1-Tube Outlet	750.000	270.000	130.600
30	A2-Shell inlet	2060.000	90.000	71.863
30	B2-Shell outlet	2060.000	270.000	71.863

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
A2-Shell inlet	150 mm	120	WNF	168.3	14.275	268.27	8.00	214	150
B2-Shell outlet	150 mm	120	WNF	168.3	14.275	270.00	8.00	214	150
M1-Man Hole	200 mm	160	WNF	219.1	23.012	...	...	226	2500
A1-Tube Inlet	300 mm	160	WNF	323.9	33.325	423.85	135.00	309	2500
B1-Tube Outlet	300 mm	160	WNF	323.9	33.325	423.85	135.00	309	2500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
A2-Shell in	SA-266 2	8.000	10.000	8.000	8.000	...
B2-Shell ou	SA-312 TP304	8.000	10.000	8.000	8.000	...
M1-Man Hole	SA-312 TP304	48.000	10.000	...	...	...
A1-Tube Inl	SA-312 TP304	135.000	10.000	120.000	100.000	...
B1-Tube Out	SA-312 TP304	135.000	10.000	120.000	100.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
A2-Shell inlet	2060.000	90.0	200.00	0.00	Main Shell
B2-Shell outlet	2060.000	270.0	200.00	0.00	Main Shell
M1-Man Hole	...	0.0	150.00	0.00	Ch Dish Head
A1-Tube Inlet	750.000	90.0	150.00	0.00	Ch Shell
B1-Tube Outlet	750.000	270.0	150.00	0.00	Ch Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 52.0mm. THK X 1180.0mm. ID X 0.0mm.	SA-240 304
1	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm. THK X 1090.0mm. ID X 50.8mm.	SA-240 304
17	BAFFLES: 1082.0mm DIA X 8.0mm THK	SA-516 70
8	TIE RODS: 25mm DIA X 2500mm	SA-193 B7M
1	TUBESHEET: 1238.0mm X 260.0mm THK	SA-965 F304
1348	TUBES: 4300.0mm X 19.0mm DIA X 2.1mm THK	SA-213 TP304
2	SADDLE: 200.0mm X 167 DEG	SA-387 11
2	BAR RING STIFFENER: 1106.0mm ID X 1327.2mm OD X 30.0mm	SA-240 304
1	CYLINDRICAL SEGMENT 135.0mm. THK X 2720.4mm. X 1500.0mm.	SA-240 304
1	CYLINDRICAL SEGMENT 135.0mm. THK X 703.9mm. X 1500.0mm.	SA-240 304
1	CYLINDRICAL SEGMENT 8.0mm. THK X 2720.4mm. X 4550.0mm.	SA-240 304
1	CYLINDRICAL SEGMENT 8.0mm. THK X 703.9mm. X 4550.0mm.	SA-240 304
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
Notes							
Main Shell [16]		-196		No			
Shell Dish En[16]		-196		No			
A2-Shell inlet[1]	B	-1	-21	No	0.495	8.000	1.00
Nozzle Flg [5]	A	-18	-48	No			
B2-Shell outl[15]		-196		No			
Warmest MDMT:		-1	-21				
Ch Dish Head [9]	!	-10		No			
Ch Shell [8]	!	-10		No			
M1-Man Hole [1]	Impact Tested Material			No			
A1-Tube Inlet [1]	!	-10		No			
B1-Tube Outlet[1]	!	-10		No			
Nozzle Flg [5]	A	-18	-48	No			
Warmest MDMT:		-10	-48				
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail		
Shell	-21.0		-10.0		Pass		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	10	bars
Channel Side Design Pressure	191	bars
Shell Side Design Temperature	150.0	°C
Channel Side Design Temperature	200.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Lethal	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	-10.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/4
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Ch Dish Head
Distance "FROM" to "TO"	0 mm.
Inside Diameter	1180 mm.
Element Thickness	46 mm.
Internal Corrosion Allowance	0 mm.
Nominal Thickness	52 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 bars
Design Temperature External Pressure	310 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	129 N./mm ²
Allowable Stress, Hydrotest	194.75 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm.
Yield Stress, Operating	144 N./mm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1090 mm.
Liquid Density	0.39940E-04 kg./cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	M1-Man Hole
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	200 mm
Nozzle Schedule	160
Nozzle Class	2500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	190.04 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element#	2/4
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	1500 mm.
Inside Diameter	1090 mm.
Element Thickness	135 mm.
Internal Corrosion Allowance	0 mm.
Nominal Thickness	135 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C



Design External Pressure 0 bars
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-240 304 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 20
 Detail Type Liquid
 Detail ID Liquid: 20
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 1090 mm.
 Liquid Density 0.39940E-04 kg./cm³

Element From Node 20
 Detail Type Nozzle
 Detail ID A1-Tube Inlet
 Dist. from "FROM" Node / Offset dist 750 mm.
 Nozzle Diameter 300 mm
 Nozzle Schedule 160
 Nozzle Class 2500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 811.77 Kgf
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-312 TP304

Element From Node 20
 Detail Type Nozzle
 Detail ID B1-Tube Outlet
 Dist. from "FROM" Node / Offset dist 750 mm.
 Nozzle Diameter 300 mm
 Nozzle Schedule 160
 Nozzle Class 2500
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 811.77 Kgf
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-312 TP304

Element From Node 20
 Detail Type Weight
 Detail ID Partition plate
 Dist. from "FROM" Node / Offset dist 850 mm.
 Miscellaneous Weight 500 Kgf
 Offset from Element Centerline 0 mm.

Element# 3/4
 Element From Node 30
 Element To Node 40
 Element Type Cylinder
 Description Main Shell
 Distance "FROM" to "TO" 4550 mm.
 Inside Diameter 1090 mm.
 Element Thickness 8 mm.
 Internal Corrosion Allowance 0 mm.
 Nominal Thickness 8 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 10 bars
 Design Temperature Internal Pressure 150 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 150 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-240 304
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node	30
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	910 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	167.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	350 mm.
Thickness of Wear Plate	8 mm.
Contact Angle, Wear Plate (degrees)	180.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	30
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	3700 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	167.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	350 mm.
Thickness of Wear Plate	8 mm.
Contact Angle, Wear Plate (degrees)	180.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1090 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Ring
Detail ID	SR, 910.0 Fr30
Dist. from "FROM" Node / Offset dist	910 mm.
Inside Diameter of Ring	1106 mm.
Thickness of Ring	30 mm.
Outside Diameter of Ring	1327.2 mm.
Material Name	SA-240 304
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Ring
Detail ID	SR, 3800.0 Fr30
Dist. from "FROM" Node / Offset dist	3700 mm.
Inside Diameter of Ring	1106 mm.
Thickness of Ring	30 mm.
Outside Diameter of Ring	1327.2 mm.
Material Name	SA-240 304
Height of Section Ring	0 mm.
Using Custom Stiffener Section	No

Element From Node	30
Detail Type	Nozzle
Detail ID	A2-Shell inlet
Dist. from "FROM" Node / Offset dist	300 mm.



Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	33.442	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	B2-Shell outlet	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	150	mm
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	38.246	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	30	
Detail Type	Weight	
Detail ID	TOP HE EMPTY	
Dist. from "FROM" Node / Offset dist	650	mm.
Miscellaneous Weight	21863	Kgf
Offset from Element Centerline	2800	mm.

Element From Node	30	
Detail Type	Weight	
Detail ID	TOP HE OPERATIN	
Dist. from "FROM" Node / Offset dist	650	mm.
Miscellaneous Weight	3937	Kgf
Offset from Element Centerline	2800	mm.

Element From Node	30	
Detail Type	Weight	
Detail ID	TOP HE TEST	
Dist. from "FROM" Node / Offset dist	650	mm.
Miscellaneous Weight	5637	Kgf
Offset from Element Centerline	2800	mm.

Element From Node	30	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	0	mm.
Force in X Direction	1250	Kgf
Force in Y Direction	0	Kgf
Force in Z Direction	370	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	

Element From Node	30	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	0	mm.
Force in X Direction	3650	Kgf
Force in Y Direction	0	Kgf
Force in Z Direction	3650	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	



Element From Node	40
Element To Node	50
Element Type	Elliptical
Description	Shell Dish End
Distance "FROM" to "TO"	50.8 mm.
Inside Diameter	1090 mm.
Element Thickness	8 mm.
Internal Corrosion Allowance	0 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 bars
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1090 mm.
Liquid Density	0.0009984 kg./cm ³



XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Ch Dish Head		...	...	...	...
Ch Shell		1500	...	1500	...
Main Shell		6310	...	4550	...
Shell Dish End		6360.8	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Ch Dish Head		191	52	...	1180	129	1
Ch Shell		191	135	...	1090	129	2
Main Shell		10.107	8	...	1090	138	3
Shell Dish End		10.107	10	...	1090	138	4

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Ch Dish Head		191	193.682	207.193	46	45.3391	1
Ch Shell		191	285.467	305.379	135	86.9805	2
Main Shell		10	20.0012	20.108	8	4.00645	3
Shell Dish End		10	22.3077	22.4145	8	3.59919	4

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

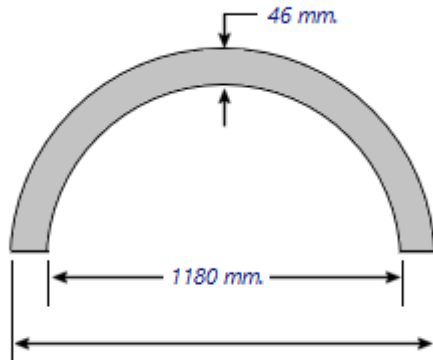
Shell Side MAWP = 14 bars
 Shell Side MAPnc = 19 bars
 Channel Side MAWP = 194 bars
 Channel Side MAPnc = 207 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Ch Dish Head

Spherical Head Design Information:					
Design Pressure	P	191.004 bars	Int. Design Temperature		200 °C
Material		SA-240 304	Ext. Design Temperature		310 °C
Length		0.000 mm.	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm.	Allowable Stress (ope)	S	129.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1180.000 mm.	Actual Stress		117.9 N./mm ²
Spec. Min. Thk t _e , or s _a		46.0000 mm.	Spec. Nominal Thk.		52.000 mm.
Surface Area		25897.0 cm ²	SG of Contents		0.040
Element Volume		430221.8 cm ³	Weight of Contents		16.9 Kg _f
Empty Weight		995.8 Kg _f	Operating Weight		1012.7 Kg _f
			Comp. All. Ope. Stress		62.5 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 14.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.375 mm.

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2 \cdot (\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 1180.0/2 \cdot (\exp(0.5 \cdot 191.004/(129.001 \cdot 1.0)) - 1) + 0.0 + 0.0$
 $= 45.3391 + 0.0 + 0.0 = 45.3391 \text{ mm.}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$
 $= 2 \cdot 129.0 \cdot 1.0 \cdot \ln((2 \cdot 46.0 + 1180.0)/1180.0) - 0.004$
 $= 193.686 - 0.004 = 193.6821 \text{ bars}$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$
 $= 2 \cdot 137.997 \cdot 1.0 \cdot \ln((2 \cdot 46.0 + 1180.0)/1180.0)$
 $= 207.1933 \text{ bars}$

Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 191.004 \cdot 1180.0^2 / (1.0 (1272.0^2 - 1180.0^2)) \\ &= 117.903 \text{ N./mm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 191.004 / (1.0 \cdot \ln((2 \cdot 46.0 + 1180.0) / 1180.0)) \\ &= 127.215 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	6.331 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	43.104 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 25897.0 / 3.14159) \\ &= 1815.850 \text{ mm.} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

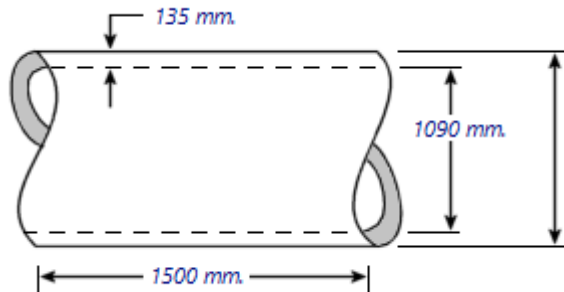
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 14.750 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.375 mm.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	191.004 bars	Int. Design Temperature		200 °C
Material		SA-240 304	Ext. Design Temperature		200 °C
Length		1500.000 mm.	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm.	Allowable Stress (ope)	S	129.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1090.000 mm.	Actual Stress		77.1 N./mm ²
Spec. Min. Thk t _e , or s _a		135.0000 mm.	Spec. Nominal Thk.		135.000 mm.
Surface Area		64088.5 cm ²	SG of Contents		0.040
Element Volume		1399948.2 cm ³	Weight of Contents		56.0 Kg
Empty Weight		6255.8 Kg	Operating Weight		6311.7 Kg
			Comp. All. Ope. Stress		79.6 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1090.0 \cdot (\exp(191.004/(129.001 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 86.9805 + 0.0 + 0.0 = 86.9805 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 129.001 \cdot 1.0 \cdot \ln((2 \cdot 135.0 + 1090.0) / 1090.0) - 0.004$$

$$= 285.472 - 0.004 = 285.4674 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 135.0 + 1090.0) / 1090.0)$$

$$= 305.3792 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 191.004 \cdot 1090.0 / (1.0 (1360.0 - 1090.0))$$

$$= 77.1136 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$\begin{aligned} &= 191.004 / (1.0 \cdot \ln((2 \cdot 135.0 + 1090.0) / 1090.0)) \\ &= 86.3124 \text{ N./mm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 11.020 %

Minimum Design Metal Temperature Results:

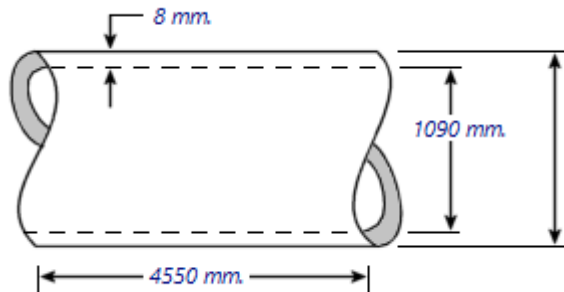
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	10.107 bars	Int. Design Temperature		150 °C
Material		SA-240 304	Ext. Design Temperature		150 °C
Length		4550.000 mm.	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1090.000 mm.	Actual Stress		68.9 N./mm ²
Spec. Min. Thk t _e , or sa		8.0000 mm.	Spec. Nominal Thk.		8.000 mm.
Surface Area		158094.4 cm ²	SG of Contents		1.000
Element Volume		2384601.5 cm ³	Weight of Contents		2383.1 Kg
Empty Weight		1007.9 Kg	Operating Weight		3391.1 Kg
			Comp. All. Ope. Stress		63.6 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1090.0 \cdot (\exp(10.107/(137.997 \cdot 1.0)) - 1) + 0.0 + 0.0$$

$$= 4.0064 + 0.0 + 0.0 = 4.0064 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 8.0 + 1090.0) / 1090.0) - 0.107$$

$$= 20.108 - 0.107 = 20.0012 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 8.0 + 1090.0) / 1090.0)$$

$$= 20.108 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 10.107 \cdot 1090.0 / (1.0 (1106.0 - 1090.0))$$

$$= 68.8569 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 10.107 / (1.0 \cdot \ln((2 \cdot 8.0 + 1090.0) / 1090.0))$$

$$= 69.3613 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 0.729 %

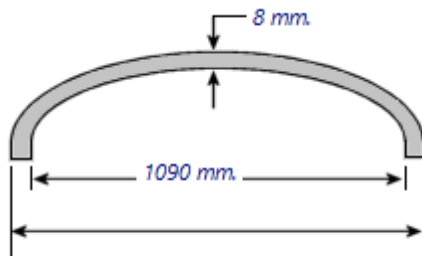
SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

This is not a Carbon Steel Material (No MDMT Calc Performed).

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell Dish End

Elliptical Head Design Information:							
Design Pressure	P	10.107	bars	Int. Design Temperature		150	°C
Material		SA-240	304	Ext. Design Temperature		150	°C
Length		50.800	mm.	External Pressure Chart		HA-1	
				UNS Number		S30400	
Int. Corr. All.	c	0.0000	mm.	Allowable Stress (ope)	S	138.0	N./mm ²
Ext. Corr. All.	ce	0.0000	mm.	Allowable Stress (amb)	Sa	138.0	N./mm ²
Inside Diameter	Di	1090.000	mm.	Actual Stress		62.2	N./mm ²
Spec. Min. Thk t,e, or sa		8.0000	mm.	Spec. Nominal Thk.		10.000	mm.
Surface Area		15127.3	cm ²	SG of Contents		1.000	
Element Volume		216960.9	cm ³	Weight of Contents		216.8	Kgf
Empty Weight		130.3	Kgf	Operating Weight		347.2	Kgf
				Comp. All. Ope. Stress		63.6	N./mm ²
				Aspect Ratio		2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.625 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.813 mm.

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:
 $t_r = 3.5992 \text{ mm.}$

$t = t_r + c_i + c_o$
 $= 3.5992 + 0.0 + 0.0$
 $= 3.5992 \text{ mm. - see below for the derivation}$

For Elliptical heads: Compute [h, k, r.] per 4.3.7.1

$h = D / (2 \cdot \text{Aspect Ratio})$
 $= 1090.0 / (2 \cdot 2.0)$
 $= 272.500 \text{ mm.}$

$k = D / (2 \cdot h)$
 $= 1090.0 / (2 \cdot 272.5)$
 $= 2.000$

$r = D (0.5/k - 0.08) + c_a$
 $= 1090.0 (0.5/2.0 - 0.08) + 0.0$
 $= 185.300 \text{ mm.}$

$$\begin{aligned} L &= D(0.44 \cdot k + 0.02) + ca \\ &= 1090.0(0.44 \cdot 2.0 + 0.02) + 0.0 \\ &= 981.000 \text{ mm.} \end{aligned}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:
 Equations 4.3.8 to 4.3.11 as appropriate
 t (corroded) and P for the calc.: $t = 3.5992 \text{ mm.}$, $P = 10.1068 \text{ bars}$

Angle [βTh]:

$$\begin{aligned} &= \arccos[(0.5 \cdot D - r)/(L - r)] \\ &= \arccos[(0.5 \cdot 1090.0 - 185.3)/(981.0 - 185.3)] \\ &= 1.1017 \text{ radians - Equation 4.3.8} \end{aligned}$$

$$\begin{aligned} \phi Th &= \sqrt{(L \cdot t) / r} \\ &= \sqrt{(981.0 \cdot 3.599) / 185.3} \\ &= 0.3207 \text{ radians - Equation 4.3.9} \end{aligned}$$

$\phi Th < \beta Th$, Thus:

$$\begin{aligned} Rth &= (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r \\ &= (0.5 \cdot 1090.0 - 185.3) / \cos(1.102 - 0.321) + 185.3 \\ &= 691.7983 \text{ mm. - Equation 4.3.10} \end{aligned}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [C1 and C2]:

$r/D > 0.08$, Thus:

$$\begin{aligned} C1 &= 0.692(r/D) + 0.605 \\ &= 0.692(185.3/1090.0) + 0.605 \\ &= 0.7226 \text{ - Equation 4.3.13} \end{aligned}$$

$$\begin{aligned} C2 &= 1.46 - 2.6(r/D) = 1.46 - 2.6(185.3/1090.0) \\ &= 1.018, \text{ Equation 4.3.15} \end{aligned}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$\begin{aligned} &= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r)) \\ &= 0.723 \cdot 186083.266 \cdot 3.599^2 / (1.018 \cdot 691.798(0.5 \cdot 691.798 - 185.3)) \\ &= 154.01 \text{ bars, Equation 4.3.16} \end{aligned}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 154.001 \text{ N./mm}^2$, Not in the creep range - 4.3.6.1(f)

$$\begin{aligned} Py &= C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1)) \\ &= 154.001 \cdot 3.599 / (1.018 \cdot 691.798(0.5 \cdot 691.798 / 185.3 - 1)) \\ &= 9.08 \text{ bars, Equation 4.3.17} \end{aligned}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 154.007/9.08 = 16.96$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$\begin{aligned} &= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) / \\ &\quad (1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py \\ &= ((0.77508 \cdot 16.96 - 0.20354 \cdot 16.96^2 + 0.019274 \cdot 16.96^3) / \\ &\quad (1 + 0.19014 \cdot 16.96 - 0.089534 \cdot 16.96^2 + 0.0093965 \cdot 16.96^3)) \cdot 9.08 \\ &= 18.16 \text{ bars, Equation 4.3.19} \end{aligned}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 18.16/1.5 = 12.11 \text{ bars - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L/t + 0.5) \\ &= 2 \cdot 137.997 \cdot 1.0 / (981.0/3.599 + 0.5) \\ &= 10.11 \text{ bars, Equation 4.3.22} \end{aligned}$$

Step 10: Compute final pressure [Pa]:

$$\begin{aligned} &= \min(Pak, Pac) = \min(12.108, 10.107) \\ &= 10.11 \text{ bars, Equation 4.3.23: (same as original design pressure)} \end{aligned}$$

Allowable pressure in Crown at given thickness [Pac]:



$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 137.997 \cdot 1.0 / (981.0 / 8.0 + 0.5) \\ &= 22.415 \text{ bars} \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:
= MaxPressure - Pliq = 22.4145 - 0.107 = 22.3077 bars

Computed Maximum Allowable Pressure New and Cold [MAPnc]:
= 22.4145 bars

Computed Membrane Stress Intensity at Design Pressure [Sm]:
= 62.2 N./mm²

Required Thickness of Straight Flange = 4.006 mm.

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	3.941 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	24.156 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:
= square root($4 \cdot \text{Head Surface Area} / 3.14159$)
= square root($4 \cdot 15127.3 / 3.14159$)
= 1387.827 mm.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

This is not a Carbon Steel Material (No MDMT Calc Performed).

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.625 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.813 mm.

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Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:
 = Test Factor • ShellSide MAWP(St/S)
 = 1.25 • 14.2(1.0)
 = 17.750 bars

Hydrotest Pressure Calculation [Channel Side]:
 = Test Factor • Channel Side MAWP(St/S)
 = 1.25 • 193.682(1.064)
 = 257.503 bars

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Ch Dish Head	171.6	194.7	0.881	257.62
Ch Shell	116.4	194.7	0.598	257.61
Main Shell	122.6	194.7	0.629	17.86
Shell Dish End	109.9	194.7	0.565	17.86

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1-Man Hole	Nozzle	117.00	110.00	1.064
A1-Tube Inlet	Nozzle	138.00	129.00	1.070
A1-Tube Inlet	Pad	138.00	129.00	1.070
B1-Tube Outlet	Nozzle	138.00	129.00	1.070
B1-Tube Outlet	Pad	138.00	129.00	1.070
A2-Shell inlet	Nozzle	165.00	146.00	1.130
A2-Shell inlet	Pad	138.00	138.00	1.000
B2-Shell outlet	Nozzle	138.00	138.00	1.000
B2-Shell outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
SR, 910.0 Fr30	138.00	138.00	1.000
SR, 3800.0 Fr30	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch Dish Head	138.00	129.00	1.070
Ch Shell	138.00	129.00	1.070
Main Shell	138.00	138.00	1.000
Shell Dish End	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	129.00	1.070



Tubesheet Material	138.00	129.00	1.070
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Minimum 1.070

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
M1-Man Hole	109.24	194.75	0.561
A1-Tube Inlet	111.80	194.75	0.574
B1-Tube Outlet	111.80	194.75	0.574
A2-Shell inlet	9.61	237.50	0.040
B2-Shell outlet	9.61	194.75	0.049

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		17.75 bars
Shell Thickness	ts	8.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1090.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		257.50 bars
Channel Thickness	tc	135.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1090.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1348
Tube Wall Thickness	et	2.1000 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	4300.00 mm.
Straight Tube Length (bet. inner tubsht faces)	L	4040.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		138.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1000 mm.
Groove Weld Leg	ag	2.1000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	512.600 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	7543.9 cm ²
Perimeter of Tube Layout	Cp	3216.8 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

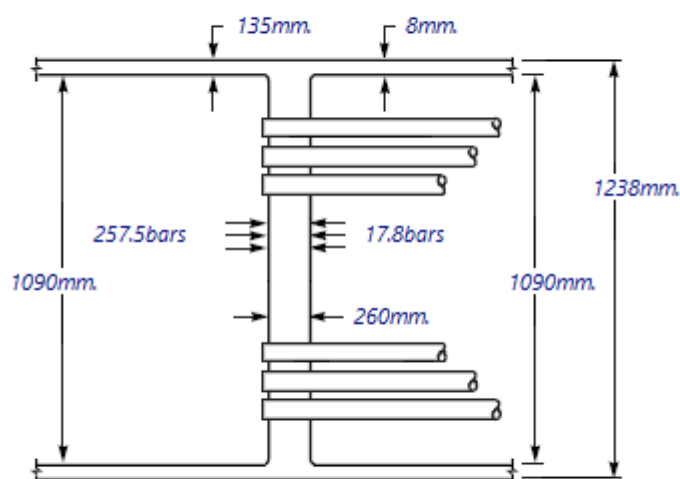
Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-965 F304
Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	260.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	1238.000 mm.

Area of the Untubed Lanes	AL	0.0 cm ²
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Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.



Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i$ Equation 4.3.1
= $0.5 \cdot 14.85 \cdot (\exp(257.503/(138.0 \cdot 1.0)) - 1) + 0$
= $1.5233 + 0$
= 1.5233 mm.

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(R_o \cdot t)}$
= $4552.6 / \sqrt{(9.525 \cdot 2.1)}$
= 1017.929

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 / (19.05/2.1)$
= 0.0606

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot E_y (t / Do)$
= $1.6 \cdot 0.061 \cdot 0.19512E+09 (2.1/19.05)$
= 2086.661 N./mm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / F_y
= $2086.7/205.0$
= 10.179

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $205.0/525.4$
= 0.390

Ratio [AE]:

= Fhe / E
= $2086.7/0.19512E+09$
= 0.010694

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
= $(\ln(0.39) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
= 0.205

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
= $205.0 (1 + 0.002) / \ln(1 + 0.002)^{0.204846}$
= 733.795 N./mm²

Curve Fitting Exponent for, Stainless Steel and Nickel Base Alloys [m2]:

= $m2f(m2f2 - R)$
= $0.75 (1.0 - 0.39)$
= 0.457

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
= $(525.4 \cdot \exp(0.457)) / (0.457^{0.457})$
= 1187.151 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
= $1.5 \cdot 0.391.5 - 0.5 \cdot 0.392.5 - 0.393.5$
= 0.281

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	262	-0.718	0.617E-05	-0.784E-06	0.154E-04	0.160E-04	0.268E+07	0.873E-01
2	131	-3.636	0.419E-06	0.417E-06	0.676E-05	-0.673E-05	0.723E+08	0.888E-02
3	197	-2.177	0.202E-05	0.188E-05	0.109E-04	-0.956E-05	0.173E+08	0.703E-03
4	164	-2.907	0.995E-06	0.981E-06	0.881E-05	-0.858E-05	0.368E+08	0.624E-02
5	180	-2.542	0.144E-05	0.139E-05	0.986E-05	-0.930E-05	0.256E+08	0.363E-02
6	188	-2.360	0.171E-05	0.163E-05	0.104E-04	-0.951E-05	0.211E+08	0.175E-02
7	192	-2.268	0.186E-05	0.175E-05	0.107E-04	-0.956E-05	0.191E+08	0.604E-03
8	194	-2.223	0.194E-05	0.182E-05	0.108E-04	-0.957E-05	0.182E+08	0.271E-04
9	193	-2.246	0.190E-05	0.178E-05	0.107E-04	-0.957E-05	0.186E+08	0.294E-03
10	194	-2.234	0.192E-05	0.180E-05	0.108E-04	-0.957E-05	0.184E+08	0.135E-03
11	194	-2.228	0.193E-05	0.181E-05	0.108E-04	-0.957E-05	0.183E+08	0.542E-04
12	194	-2.226	0.193E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.136E-04
13	194	-2.224	0.194E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.671E-05
14	194	-2.225	0.194E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.347E-05
15	194	-2.225	0.194E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.162E-05
16	194	-2.225	0.194E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.927E-06
17	194	-2.225	0.194E-05	0.181E-05	0.108E-04	-0.957E-05	0.182E+08	0.346E-06

Factor of Safety [FS]:
 Since Fic > 0.55 Fy and Fic < Fy
 $= 2.407 - 0.741 \cdot \text{Fic} / \text{Fy}$
 $= 2.407 - 0.741 \cdot 194.89/205.0$
 $= 1.703$

Allowable Circumferential Compressive Stress [Fha]:
 $= \text{Fic} / \text{FS}$
 $= 194.888 / 1.703$
 $= 114.469 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:
 $= 2 \cdot \text{Fha} (t / \text{Do})$
 $= 2 \cdot 1144.625 (2.1/19.05)$
 $= 252.358 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.5233 mm.
Allowable Internal Pressure at Corroded thickness	343.69 bars
Required Internal Design Pressure	257.50 bars
Allowable External Pressure at Corroded thickness	252.36 bars
Required External Design Pressure	17.75 bars
Required Thickness due to Shell Side pressure	0.4414 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot \text{ro} + \text{dt}$
 $= 2 \cdot 512.6 + 19.05$
 $= 1044.250 \text{ mm.}$

Basic ligament efficiency for shear [μ]:
 $= (p - \text{dt}) / p$
 $= (25.4 - 19.05) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:
 $= \text{Ds} / \text{Do} \text{ (Configurations a, b, c)}$



$$= 1090.0/1044.2499$$

$$= 1.0438$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1090.0/1044.2499$$

$$= 1.0438$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/260.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t(E_t T/E) \cdot (StT/S) \cdot (\rho); dt - 2t_t)$$

$$= \max(19.05 - 2 \cdot 2.1 (0.19512E+09/1.9512E+09) \cdot$$

$$(137/137) \cdot (0); 19.05 - 2 \cdot 2.1)$$

$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 25.4 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1044.25 \cdot 25.4) / (\pi \cdot 1044.25^2)) }$$

$$= 25.4000 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19.05) / 25.4$$

$$= 0.2500$$

$$h/p = 10.236220 ; \mu^* = 0.250000 ; E^* = 42264436 \text{ KPa.}$$

$$E^*/E = 0.216611 ; \nu^* = 0.396457$$

Note: As h/p (10.236) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{ ((1090.0 + 8.0) 8.0) }$$

$$= 0.0194 \text{ 1/mm.}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.019 \cdot 1951171 \cdot 8.0^3 / (6(1 - 0.31^2))$$

$$= 3566753.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1090.0 + 8.0) \cdot 3566753 / 260.0^3 \cdot (1 + 260.0 \cdot 0.019 + (260.0 \cdot 0.019)^2 / 2)$$

$$= 25010.3 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1090.0^2 / (4 \cdot 0.19512E+09 \cdot 8.0) \cdot (1 - 1090.0 \cdot 0.31 / (2(1090.0 + 8.0)))$$

$$= 0.16100E+00 \text{ mm./N./mm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0438 \cdot 3566753 \cdot 0.0194 \cdot 0.160997 (1 + 260.0 \cdot 0.0194)$$

$$= 7004.1709 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{ ((1090.0 + 135.0) 135.0) }$$

$$= 0.0045 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$



$$= 0.004 \cdot 1951171 \cdot 135.0^3 / (6(1 - 0.31^2))$$

$$= 3950168064.0000 \text{ bars} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1090.0 + 135.0 / \cos(0)) \cdot 0.39502E+1 / 260.0^3 \cdot (1.0 + 260.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (260.0 \cdot 0.004)^2 / 2 \cdot 1.0)$$

$$= 4680502.0 \text{ bars}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1090.0^2 / (4 \cdot 0.19512E+09 \cdot 135.0) \cdot (1 - 1090.0 \cdot 0.31 / (2(1090.0 + 135.0)))$$

$$= 0.97205E-02 \text{ mm}^2 / \text{N} \cdot \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= p_c \cdot k_c [\delta c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.044 \cdot 0.39502E+1 [0.00972 (1.0 + 0.004 \cdot 1.0 \cdot 260.0) - 0 (1.0 / 260.0 + 1.0 \cdot 0.004 / 2)]$$

$$= 38641.2227 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1238.0 / 1044.2499$$

$$= 1.1855$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 42264436 \cdot (25010.28 + 4680502 + 0.19512E+09 \cdot \ln(1.19))$$

$$= 7.1938$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1044.25^2 / 16 \cdot [(1.044 - 1) \cdot (1.044^2 + 1) \cdot 17.75 - (1.044 - 1) \cdot (1.044^2 + 1) \cdot 257.503]$$

$$= -1495857.8750 \text{ bars} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot P_t - \omega s \cdot P_s$$

$$= -1495857.8 + 38641.223 \cdot 257.503 - 7004.171 \cdot 17.75$$

$$= 8330033.5000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (8330033) - 1044.25^2 / 32 \cdot 7.194 (17.75 - 257.5) / (1 + 7.19)$$

$$= 8189523.5000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 8189523 + 1044.25^2 / 64 \cdot (3 + 0.396) \cdot (17.75 - 257.5)$$

$$= -5684989.0000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|8189523|; |-5684989.0|)$$

$$= 8189523.5000 \text{ bars} \cdot \text{mm}^2$$



ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 8189523 / (0.25 \cdot (260.0 - 6.0)^2) \\ &= 304.6686 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 205.0 \\ &= 389.49 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 8126393 / (0.25 \cdot (229.9521 - 6.0)^2) \\ &= 388.8875 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 229.9521 + 0 + 0 \\ &= 229.9521 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(229.9521; 144.3625) \\ &= 229.9521 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |17.75 - 257.5| \\ &= 239.752 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.25 \cdot 260.0 / 1044.25 \cdot 0.8 \cdot 0.95 \cdot 204.996 \\ &= 38.791 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.25 \cdot 155.797) \cdot (4 \cdot 7543.9 / 3216.83) \cdot |17.75 - 257.5| + 0 \\ &= 144.3625 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 8.000mm.

Min. Shell len. adjacent to the tubesheet is: 168.086 mm.

Note: For a given Channel thickness of: 135.000 mm.

Min. Channel len. adjacent to the tubesheet is: 690.482mm.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.1) \cdot 2.1 \\ &= 1.118 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.1 (19.05 - 2.1) \cdot 194.746 \\ &= 2220.648 \text{ Kg} \end{aligned}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.1 (19.05 + 0.67 \cdot 2.1) 194.75$$

$$= 2220.6484 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194.746/194.746$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.1 (19.05 + 0.67 \cdot 2.1) 194.75$$

$$= 1474.0587 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.1 \cdot (19.05 - 2.1) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.552 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2759 mm.

Minimum Required Groove Weld Leg [agr]: 1.2759 mm.

Tube-Tubesheet Jt allowable, 2220.65 is $\geq$ tube strength 2220.65 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	304.7 $\leq$	389.5 N./mm ²	Ok
Tubesheet shear stress	86.5 $\leq$	155.8 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	229.952	260.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.276	2.100	Ok
Tube-Tubesheet Groove Weld Leg :	1.276	2.100	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear Allwd	Case Type	Pass/ Fail
D1uc	237.313	...	324	389	93	Ps+Pt D1	Ok
D2uc	66.884	...	20	389	6	Ps+Pt D2	Ok
D3uc	229.952	...	305	389	87	Ps+Pt D3	Ok
Max:	237.3132	... mm.	0.833	0.596	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.							
Case#	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	292	...	292	Ok
D2uc	...	...	...	292	...	292	Ok
D3uc	...	...	...	292	...	292	Ok
Max	...	...	...	...	...	...	...

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	237.313	260.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	1.523	2.100	Ok
Tube-tubesheet fillet weld leg :	1.276	2.100	Ok
Tube-tubesheet groove weld leg :	1.276	2.100	Ok

Given Shell Thickness: 8.0000 mm.
 Given Channel Thickness: 135.0000 mm.

Min Shell length of thk, (8.000) adj. to tubesheet: 168.086 mm.

Min Channel length of thk, (135.000) adj. to tubesheet: 690.482 mm.

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1272	46	No Calc	No Calc
20	Tsht	1696.64	1360	135	No Calc	No Calc
Tsht	30	No Calc	...	135	No Calc	No Calc
30	Ring	910.031	1106	8	No Calc	No Calc
Ring	Ring	2790	1106	8	No Calc	No Calc
Ring	40	991.633	1106	8	No Calc	No Calc
40	50	No Calc	1106	8	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	46	No Calc	...	89.3235
20	Tsht	135	No Calc	...	195.961
Tsht	30	135	No Calc	...	No Calc
30	Ring	8	No Calc	...	5.43777
Ring	Ring	8	No Calc	...	2.94253
Ring	40	8	No Calc	...	5.25125
40	50	8	No Calc	...	5.4592

Minimum

3

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Tsht	1696.64	No Calc	No Calc	No Calc
Tsht	30	No Calc	No Calc	No Calc	No Calc
30	Ring	910.031	No Calc	No Calc	No Calc
Ring	Ring	2790	No Calc	No Calc	No Calc
Ring	40	991.633	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	995.8	430222	995.8	430222	99.58
20	30	6255.77	1399948	6255.77	1399948	625.577
30	40	1007.91	2384601	1007.91	2384601	100.791
40	50	130.343	216961	130.343	216961	13.0343
Total		8389	4431732.50	8389	4431732.50	838



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	16.9152	-196.667	45	...	Liquid: 10
10	Nozl	209.04	-590	...	...	M1-Man Hole
20	Liqd	55.9694	750	...	...	Liquid: 20
20	Nozl	892.944	750	673.6	...	A1-Tube Inlet
20	Nozl	892.944	750	-673.6	...	B1-Tube Outlet
20	Wght	500	850	...	...	Partition plate
30	Sadl	421.674	910	914.5	...	Left Saddle
30	Sadl	421.674	3700	914.5	...	New Saddle
30	Liqd	2383.15	2275	...	...	Liquid: 30
30	Ring	111.979	910	...	...	SR, 910.0 Fr30
30	Ring	111.979	3700	...	...	SR, 3800.0 Fr30
30	Nozl	36.7858	300	614.863	...	A2-Shell inlet
30	Nozl	42.0702	300	-614.863	...	B2-Shell outlet
30	Wght	21863	650	2800	...	TOP HE EMPTY
30	Wght	3937	650	2800	...	TOP HE OPERATIN
30	Wght	5637	650	2800	...	TOP HE TEST
30	Forc	...	...	...	...	TOP HE FORCES
30	Forc	...	...	...	...	TOP HE FORCES
40	Liqd	216.829	141.633	...	...	Liquid: 40
20	FTsh	1881.47	1630	...	...	
20	Tube	6605.42	3650	...	...	Tubes
30	Baf1	40.8833	460	...	...	Baffles
30	Baf1	40.8833	700	...	...	Baffles
30	Baf1	40.8833	940	...	...	Baffles
30	Baf1	40.8833	1180	...	...	Baffles
30	Baf1	40.8833	1420	...	...	Baffles
30	Baf1	40.8833	1660	...	...	Baffles
30	Baf1	40.8833	1900	...	...	Baffles
30	Baf1	40.8833	2140	...	...	Baffles
30	Baf1	40.8833	2380	...	...	Baffles
30	Baf1	40.8833	2620	...	...	Baffles
30	Baf1	40.8833	2860	...	...	Baffles
30	Baf1	40.8833	3100	...	...	Baffles
30	Baf1	40.8833	3340	...	...	Baffles
30	Baf1	40.8833	3580	...	...	Baffles
30	Baf1	40.8833	3820	...	...	Baffles
30	Baf1	40.8833	4060	...	...	Baffles
30	Baf1	41.8812	4300	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	843.3
Liquid	2672.9
Stiffeners	224.0

Nozzles 2073.8
 Weights 31937.0
 Exchanger Components 9182.9
 Liquid in Tubes 1191.4

Sum of the Detail Weights 48125.2 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	9228.8	9228.8	9228.8	9228.8	9228.8	9228.8
Saddles	843.3	843.3	843.3	843.3	843.3	843.3
Stif. Rings	224.0	224.0	224.0	224.0	224.0	224.0
Nozzles	2073.8	2073.8	2073.8	2073.8	2073.8	2073.8
Wld Weights	22363.0	22363.0	22363.0	22363.0	22363.0	22363.0
Exchanger	9182.9	9182.9	9182.9	9182.9	9182.9	9182.9
Test Weights	...	5637.0	...	...	...	...
Ope Weights	...	...	...	...	...	3937.0
Ope. Liquid	...	...	...	...	...	2672.9
Tube Ope Lqd	...	...	...	...	...	1191.4
Test Liquid	...	4429.0	...	...	...	...
Tube Tst Lqd	...	1191.4	...	...	...	...
Totals	43915.8	55173.2	43915.8	43915.8	43915.8	51717.1

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	43915.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	55173.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	43915.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	43915.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	43915.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	51717.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	51717.1 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1192126.8 cm ³
Weight of Ope Liq in tubes :	1191.4 kg.
Weight of Water in tubes :	1191.4 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	25897	21871.8
20	30	64088.5	51365
30	40	158094	155807
40	50	15127.3	14618.4

Total	263207.2	243662.5 cm ²
Total	283.3	262.3 ft ²



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	50%	Max Pressure DNV
M1-Man Hole	285.70	414.30	200	2500	GR 2.1	...	...	...	...
A1-Tube Inlet	285.70	414.30	200	2500	GR 2.1	14.741	85.705	85.705	85.705
B1-Tube Outlet	285.70	414.30	200	2500	GR 2.1	14.741	85.705	85.705	85.705
A2-Shell inlet	15.80	19.60	150	150	GR 1.1	3.014	15.800	15.800	15.800
B2-Shell outle	14.20	19.10	150	150	GR 2.1	3.014	14.200	14.200	14.200

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 14.200 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 19.100 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 285.705 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 414.300 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
A1-Tube Inlet	342.900	0.50	15	13706	193.682	285.705	Pass
B1-Tube Outlet	342.900	0.50	15	13706	193.682	285.705	Pass
A2-Shell inlet	196.088	1.20	1	918	14.200	15.800	Pass
B2-Shell outlet	196.088	1.20	1	918	14.200	14.200	Pass

Evaluating Flange on Nozzle [A1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(15) + 4 \cdot \text{abs}(13706) \cdot 342.900 \leq \pi \cdot 342.900^3 \cdot ((285.705 - 193.682) + 0.500 \cdot 285.705)$$

$$= 0.19041E+05 \leq 0.30338E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(285.705, 0.500 \cdot 285.705 + 285.705 - 19041.2 / (\pi \cdot 342.900^3))$$

$$= \min(285.705, 413.815)$$

$$= 285.705 \text{ bars}$$

Evaluating Flange on Nozzle [B1-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(15) + 4 \cdot \text{abs}(13706) \cdot 342.900 \leq \pi \cdot 342.900^3 \cdot ((285.705 - 193.682) + 0.500 \cdot 285.705)$$

$$= 0.19041E+05 \leq 0.30338E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle B1-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(285.705, 0.500 \cdot 285.705 + 285.705 - 19041.2 / (\pi \cdot 342.900^3))$$

$$= \min(285.705, 413.815)$$

$$= 285.705 \text{ bars}$$

Evaluating Flange on Nozzle [A2-Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1) + 4 \cdot \text{abs}(918) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((15.800 - 14.200) + 1.200 \cdot 15.800)$$

$$= 0.72799E+03 \leq 0.49664E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A2-Shell inlet [Pmax]:
= $\min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
= $\min(15.800, 1.200 \cdot 15.800 + 15.800 - 728.0 / (\pi \cdot 196.088^3))$
= $\min(15.800, 31.747)$
= 15.800 bars

Evaluating Flange on Nozzle [B2-Shell outlet]:
= $16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
= $16(1) + 4 \cdot \text{abs}(918) \cdot 196.088 \leq \pi \cdot 196.088^3 [(14.200 - 14.200) + 1.200 \cdot 14.200]$
= $0.72799E+03 \leq 0.41161E+04$ Check Passes

Maximum Pressure for Flange on Nozzle B2-Shell outlet [Pmax]:
= $\min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
= $\min(14.200, 1.200 \cdot 14.200 + 14.200 - 728.0 / (\pi \cdot 196.088^3))$
= $\min(14.200, 28.227)$
= 14.200 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotect	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.987) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.919)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.583
Structure Height to Diameter ratio	5.983

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1300.0 + 0.0 = 1300.0$ mm.
 $= 4.265$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (4.265 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.583 \cdot 6.84$$

$$= 112.4 \text{ lbs. } [51.0] \text{ Kgf}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Ch Dish Head	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Ch Shell	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Main Shell	1300.0	0.000	0.000	0.000	1.035	1.000	161.849
Shell Dish End	1300.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	1300.00	161.85	0.00
TOP HE EMPTY	1.00	4100.00	168.34	0.02
TOP HE OPERAT	1.00	4100.00	168.34	0.02
TOP HE TEST	96776.00	4100.00	168.34	1629.09

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1300	1526.4	6354.27	161.849	56.0653
20	30	1300	1632	24480	161.849	215.993
30	40	1300	1327.2	60387.6	161.849	2324.85
40	50	1300	1327.2	3110.78	161.849	27.4472

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(6.20080.75)$
 = 0.079 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.079$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 50874 \cdot 1.5$
 = 4975.450 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 50874$
 = 2211.3 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	590	8478.95	885.525
20	30	545	8478.95	817.985
30	Sad1	545	8478.95	817.985
Sad1	40	545	8478.95	817.985
30	40	545	8478.95	817.985
40	50	545	8478.95	817.985



The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4065.000 mm.
Center of Gravity of the Liquid	4135.475 mm.
Center of Gravity of the Stiffening Rings	4065.000 mm.
Center of Gravity of the Nozzles	664.740 mm.
Center of Gravity of the Added Weights (Operating)	2380.342 mm.
Center of Gravity of the Added Weights (Empty)	2375.121 mm.
Center of Gravity of the Tubesheet(s)	1630.000 mm.
Center of Gravity of the Tubes	3650.000 mm.
Center of Gravity of the Baffles	4142.752 mm.
Center of Gravity of Bare Shell New and Cold	1108.219 mm.
Center of Gravity of Bare Shell Corroded	1108.219 mm.
Vessel CG in the Operating Condition	2397.851 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2257.034 mm.
Vessel CG in the Test Condition	2329.730 mm.

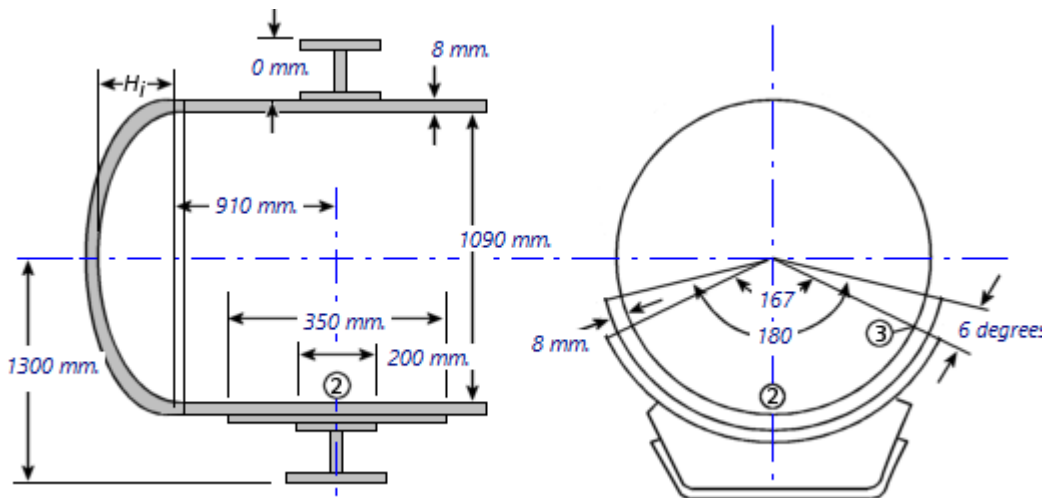
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	549.00	mm.
Shell Thickness used in this Case	t	8.000	mm.
Stiffened Vessel Length per 4.15.6	L	4600.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	910.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	350.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	8.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	613.00	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		129.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	84799.59	Kgf
Pressure used in Analysis	P 10.053	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		27.93	138.00
Long. Stress at Bottom of Midspan		41.07	138.00
Long. Stress at Top of Saddles		67.10	138.00
Long. Stress at Bottom of Saddles		1.89	138.00
Tangential Shear in Shell		30.93	82.08
Circ. Compressive Stress in Shell		12.17	138.00
Stiffener Circ. Stress at Shell		29.97	172.50
Stiffener Circ. Stress at Tip		168.79	172.50



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(2624.4/2 + 5331) \cdot 1300.0/1114.7876$
 $= 23241.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(246.3, 4900) \cdot 1300.0/2790.0$
 $= 2283.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(4975.45, 4900) \cdot 1300.0/2790.0$
 $= 2318.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(4975/2 + 5331) \cdot 1300.0/1114.7876$
 $= 27353.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 57446 + \max(2283, 23241, 2318, 27354)$
 $= 84799.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.106 \cdot 1.2)^2)$
 $= 2415.358 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	85221.27	Kgf
Transverse Shear Load	7818.91	Kgf
Longitudinal Shear Load	4975.45	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(1090.0/2 + 8.0 + 8.0)\sin(167.0/2), 2 \cdot 549.0)$
 $= 1114.788 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0229	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$
 $= -84800 \cdot 910.0 [1 - (1 - 910.0/4600.8 + (549.0^2 - 613.0^2)/(2 \cdot 910.0 \cdot 4600.8))/(1 + (4 \cdot 613.0)/(3 \cdot 4600.8))]$
 $= -25184.0 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 \{ (1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2)) / (1 + (4\text{hm})/(3L)) - 4a/L \}$
 $= 84800 \cdot 4601/4 \{ (1 + 2(549^2 - 613^2)/(4601^2)) / (1 + (4 \cdot 613)/(3 \cdot 4601)) - 4 \cdot 910/4601 \}$
 $= 5073.5 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 t)$
 $= 10.053 \cdot 549.0/(2 \cdot 8.0) - 5073.5/(\pi \cdot 549.0^2 \cdot 8.0)$
 $= 27.93 \text{ N/mm}^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.053 \cdot 549.0 / (2 \cdot 8.0) + 5073.5 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 41.07 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.053 \cdot 549.0 / (2 \cdot 8.0) - 25184.0 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 67.10 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 10.053 \cdot 549.0 / (2 \cdot 8.0) + 25184.0 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 1.89 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 84800(4600.8 - 2 \cdot 910.0) / (4600.8 + (4 \cdot 613.0 / 3))$$

$$= 43522.5 \text{ Kg}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 43522.5 / (\pi \cdot 549.0 \cdot 8.0)$$

$$= 30.93 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{549.0 \cdot 8.0}$$

$$= 51.692 \text{ mm.}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.642 \cdot 84800 \cdot 0.1 / 43.85$$

$$= -12.17 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.2738 \cdot 84800 / 43.85 + 0.0229 \cdot 84800 \cdot 549.0 \cdot 48.87 / 622.72$$

$$= 29.97 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.2738 \cdot 84800 / 43.85 - 0.0229 \cdot 84800 \cdot 549.0 \cdot 69.73 / 622.72$$

$$= -168.79 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2790.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000017 \cdot 2790.0 (150.0 - 21.1)$$

$$= 5.959 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	30.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	244.4	N./mm ²
Height of Web at Center	Hw,c	570.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(545.0 \cdot 8.0)} + 350.0)^3) / 12$
 $= 1.933 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 1.933 + 36.241 (376.426 - 4.0)^2$
 $= 50268.043 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	453.0	8.0	4.0	36.2	14496.2	0.503E+05
Wearplate	350.0	8.0	12.0	28.0	33600.0	0.372E+05
Web	20.0	709.0	370.5	141.8	5253689.5	0.594E+05
BasePlate	225.0	30.0	740.0	67.5	4995000.0	0.893E+05
Totals	...	...	...	273.5	10296785.0	0.236E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 10296785 / 273.541$
 $= 376.426 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 167.0 / 2$
 $= 96.5$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.684 + \sin(96.5) \cos(96.5))$
 $= 0.2924$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.292 \cdot 84799.594$
 $= 24793.0645 \text{ Kg}$

Tension Stress, $St = (Fh / As) = 10.2462 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 146.6399 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1300.0 - 545.0 \cdot \sin(167.0/2) / 1.4574$
 $= 928.437 \text{ mm.}$

Bending Moment, $M = Fh \cdot d = 23019.2656 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 35.9787 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 162.9332 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:
 $= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(84800 + 422) \cdot 225.0) / (4 \cdot 1200.0 \cdot 162.933))}$
 $= 26.857 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 167.0/2) (545.0 + 8.0 + 8.0)$
 $= 1114.788 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1114.7876 / (4 - 1)$
 $= 371.596 \text{ mm.}$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot Bpwid / 2 \\ &= 371.5959 \cdot 225.0 / 2 \\ &= 418.045 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth) \\ &= 85221.258 / (1200.0 \cdot 225.0) \\ &= 31.563 \text{ Kg/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 418.0 \cdot 31.56 \\ &= 13194.946 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= Rib \text{ Area} + Web \text{ Area} \\ &= 36.0 + 37.16 \\ &= 73.160 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 13194.9 / 73.1596 \\ &= 17.687 \text{ N/mm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	200.0	...	40.0	...	0.133E+04

Rib dimension [D]:

$$\begin{aligned} &= Saddle \text{ Width} - Web \text{ Thickness} \\ &= 200.0 - 20.0 \\ &= 180.000 \text{ mm.} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0 / 73.16 \\ &= 0.000 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= Saddle \text{ Width} / 2 \\ &= 200.0 / 2 \\ &= 100.000 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) } \\ &= \sqrt{ (1333.3 / 73.16) } \\ &= 42.691 \text{ mm.} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk \\ &= 1300.0 - \cos(167.0 / 2) (545.0 + 8.0 + 8.0) - 30.0 \\ &= 1206.493 \text{ mm.} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) } \\ &= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 244.4) } \\ &= 127.081 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 1206.493 / 42.691 \\ &= 28.261 \end{aligned}$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 4975.4 / (2 \cdot 1200.0) \cdot 371.596 \cdot 1206.49 / 2$$

$$= 464.724 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (28.2612 < 127.0809) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (28.26)^2 / (2 \cdot 127.08^2)) 244 / (5/3 + 3 \cdot (28.26) / (8 \cdot 127.08) - (28.26^3) / (8 \cdot 127.08^3))$$

$$= 136.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 17.69 / 136.31 + (464.72 \cdot 100.0 / 13333333) / 162.93$$

$$= 0.340$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	180.0	0.0	36.0	0.0	0.133E+04
Web	371.6	20.0	0.0	74.3	0.0	24.8
Totals	...	...	...	110.3	...	0.136E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 110.319$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1300.0 - \sqrt{(561.0 + 8.0)^2 - (371.596/2)^2} - 30.0$$

$$= 740.661 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1356.8 / 110.319}$$

$$= 35.069 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 740.661 / 35.069$$

$$= 21.120$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 4975.449 / (2 \cdot 1200.0)$$

$$= 2.073 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.073 \cdot 371.596 \cdot 740.661$$

$$= 570.584 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 570.584 / 135677.312$$

$$= 41.241 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (21.1199 < 127.0809) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.12)^2 / (2 \cdot 127.08^2)) 244 / (5/3 + 3 \cdot (21.12) / (8 \cdot 127.08) - (21.12^3) / (8 \cdot 127.08^3))$$

$$= 139.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 23.48/139.45 + (570.58 \cdot 100.0/1356.772)/162.93$$

$$= 0.421$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.38 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BlArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	57867.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2318.3 Kg
Maximum Transverse Force	Ft	7818.9 Kg
Maximum Longitudinal Force	Fl	4975.4 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (Bolt Area \cdot Number of Bolts)$$

$$= 4975.4 / (4.867 \cdot 2)$$

$$= 50.1 \text{ N./mm}^2. \text{ Must be less than } 103.4 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 7818.9 / (4.867 \cdot 2)$$

$$= 78.8 \text{ N./mm}^2. \text{ Must be less than } 103.4 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (BoltShearAllowable \cdot Nbolts)$$

$$= 4975.45 / (103.43 \cdot 2.0)$$

$$= 2.3589 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1300.0 \cdot 7818.91 + 0.0$$

$$= 10164.78 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 10164.78 / 57867.42$$

$$= 175.65 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 2.3589, 0.0]$$

$$= 2.3589 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	549.00 mm.
Shell Thickness used in this Case	t	8.000 mm.
Stiffened Vessel Length per 4.15.6	L	4600.80 mm.



Distance from Saddle to Vessel tangent	a1 or a	910.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	350.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	8.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	276.50	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		19018.88	Kgf
Pressure used in Analysis	P	10.053	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	30.13	138.00
Long. Stress at Bottom of Midspan	38.87	138.00
Long. Stress at Top of Saddles	39.71	138.00
Long. Stress at Bottom of Saddles	29.29	138.00
Tangential Shear in Shell	7.56	82.08
Circ. Compressive Stress in Shell	2.73	138.00
Stiffener Circ. Stress at Shell	6.72	172.50
Stiffener Circ. Stress at Tip	37.86	172.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(2624.4/2 + 1311) \cdot 1300.0/1114.7876$
 $= 9177.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(246.3, 4900) \cdot 1300.0/2790.0$
 $= 2283.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,\text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(4975.45, 4900) \cdot 1300.0/2790.0$
 $= 2318.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(4975/2 + 1311) \cdot 1300.0/1114.7876$
 $= 13290.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 5729 + \max(2283, 9178, 2318, 13290)$
 $= 19018.9 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.106 \cdot 1.2)^2)$
 $= 2415.358 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	19440.55	Kgf
Transverse Shear Load	3798.91	Kgf
Longitudinal Shear Load	4975.45	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1090.0/2 + 8.0 + 8.0)\sin(167.0/2), 2 \cdot 549.0)$$

$$= 1114.788 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0229	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -19019 \cdot 910.0 [1 - (1 - 910.0/4600.8 + (549.0^2 - 276.5^2)/(2 \cdot 910.0 \cdot 4600.8))/(1 + (4 \cdot 276.5)/(3 \cdot 4600.8))]$$

$$= -4022.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 19019 \cdot 4601/4((1 + 2(549^2 - 277^2)/(4601^2))/(1 + (4 \cdot 277)/(3 \cdot 4601)) - 4 \cdot 910/4601)$$

$$= 3376.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 10.053 \cdot 549.0/(2 \cdot 8.0) - 3376.0/(\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 30.13 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 10.053 \cdot 549.0/(2 \cdot 8.0) + 3376.0/(\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 38.87 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 10.053 \cdot 549.0/(2 \cdot 8.0) - 4022.8/(\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 39.71 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 t)$$

$$= 10.053 \cdot 549.0/(2 \cdot 8.0) + 4022.8/(\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 29.29 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 19019(4600.8 - 2 \cdot 910.0)/(4600.8 + (4 \cdot 276.5/3))$$

$$= 10642.5 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 10642.53/(\pi \cdot 549.0 \cdot 8.0)$$

$$= 7.56 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(549.0 \cdot 8.0)}$$

$$= 51.692 \text{ mm.}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6 *]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.642 \cdot 19019 \cdot 0.1/43.85$$

$$= -2.73 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8 *]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.2738 \cdot 19019/43.85 + 0.0229 \cdot 19019 \cdot 549.0 \cdot 48.87/622.72$$

$$= 6.72 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9 *]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.2738 \cdot 19019/43.85 - 0.0229 \cdot 19019 \cdot 549.0 \cdot 69.73/622.72$$

$$= -37.86 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	30.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	244.4	N./mm ²
Height of Web at Center	Hw,c	570.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3)/12$$

$$= ((1.56 \cdot \sqrt{(545.0 \cdot 8.0) + 350.0})^3)/12$$

$$= 1.933 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1.933 + 36.241(376.426 - 4.0)^2$$

$$= 50268.043 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	453.0	8.0	4.0	36.2	14496.2	0.503E+05
Wearplate	350.0	8.0	12.0	28.0	33600.0	0.372E+05
Web	20.0	709.0	370.5	141.8	5253689.5	0.594E+05
BasePlate	225.0	30.0	740.0	67.5	4995000.0	0.893E+05
Totals	...	...	...	273.5	10296785.0	0.236E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10296785/273.541$$

$$= 376.426 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 167.0/2$$

$$= 96.5$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2)/(\pi - 1.684 + \sin(96.5)\cos(96.5))$$

$$= 0.2924$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.292 \cdot 19018.877$$

= 5560.5952 Kgf

Tension Stress, $St = (Fh/As)$ = 2.2980 N./mm²
 Allowed Stress, $Sa = 0.6 * Yield Str$ = 146.6399 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$
 $= 1300.0 - 545.0 \cdot \sin(167.0/2)/1.4574$
 $= 928.437 \text{ mm.}$

Bending Moment, $M = Fh * d$ = 5162.7671 Kg-m.

Bending Stress, $Sb = (M * C1 / I)$ = 8.0693 N./mm²
 Allowed Stress, $Sa = 2/3 * Yield Str$ = 162.9332 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + SaddleWt)BasePlateWidth / (4 * BasePlateLength * AllStress))}$
 $= \sqrt{(3(19019 + 422)225.0 / (4 * 1200.0 * 162.933))}$
 $= 12.828 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)$
 $= 2 * \cos(90 - 167.0/2) (545.0 + 8.0 + 8.0)$
 $= 1114.788 \text{ mm.}$

Distance between Ribs [e]:
 $= Web Length / (Nribs - 1)$
 $= 1114.7876 / (4 - 1)$
 $= 371.596 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e * Bpwid / 2$
 $= 371.5959 * 225.0/2$
 $= 418.045 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + SaddleWt) / (BasePlateLength * BasePlateWidth)$
 $= 19440.551 / (1200.0 * 225.0)$
 $= 7.200 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 418.0 * 7.2$
 $= 3010.012 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= Rib Area + Web Area$
 $= 36.0 + 37.16$
 $= 73.160 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 3010.0/73.1596$
 $= 4.035 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	200.0	...	40.0	...	0.133E+04

Rib dimension [D]:
 $= Saddle Width - Web Thickness$
 $= 200.0 - 20.0$
 $= 180.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/73.16$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1333.3/73.16)}$$

$$= 42.691 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300.0 - \cos(167.0/2) (545.0 + 8.0 + 8.0) - 30.0$$

$$= 1206.493 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/244.4)}$$

$$= 127.081$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1206.493/42.691$$

$$= 28.261$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 4975.4 / (2 \cdot 1200.0) \cdot 371.596 \cdot 1206.49/2$$

$$= 464.724 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($28.2612 < 127.0809$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (28.26)^2 / (2 \cdot 127.08^2)) 244 /$$

$$(5/3 + 3 \cdot (28.26) / (8 \cdot 127.08) - (28.26^3) / (8 \cdot 127.08^3))$$

$$= 136.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.03/136.31 + (464.72 \cdot 100.0/13333333) / 162.93$$

$$= 0.239$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	180.0	0.0	36.0	0.0	0.133E+04
Web	371.6	20.0	0.0	74.3	0.0	24.8
Totals	...	...	...	110.3	...	0.136E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/110.319$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1300.0 - \sqrt{(561.0 + 8.0)^2 - (371.596/2)^2} - 30.0$$

$$= 740.661 \text{ mm.}$$

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1356.8/110.319}$
 $= 35.069 \text{ mm.}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 740.661/35.069$
 $= 21.120$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 4975.449/(2 \cdot 1200.0)$
 $= 2.073 \text{ Kg/mm.}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 2.073 \cdot 371.596 \cdot 740.661$
 $= 570.584 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 570.584/135677.312$
 $= 41.241 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ (21.1199 < 127.0809) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (21.12)^2 / (2 \cdot 127.08^2)) 244 /$
 $(5/3 + 3 \cdot (21.12) / (8 \cdot 127.08) - (21.12^3) / (8 \cdot 127.08^3))$
 $= 139.4 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 5.36/139.45 + (570.58 \cdot 100.0/1356.772) / 162.93$
 $= 0.292$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.38 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	148.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	6150.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2318.3 Kg
Maximum Transverse Force	Ft	3798.9 Kg
Maximum Longitudinal Force	Fl	4975.4 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 4975.4 / (4.867 \cdot 2)$
 $= 50.1 \text{ N./mm}^2$. Must be less than 103.4 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 3798.9 / (4.867 \cdot 2)$
 $= 38.3 \text{ N./mm}^2$. Must be less than 103.4 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:



$$\begin{aligned} &= F / (\text{BoltShearAllowable} \cdot \text{Nbols}) \\ &= 4975.45 / (103.43 \cdot 2.0) \\ &= 2.3589 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 1300.0 \cdot 3798.91 + 0.0 \\ &= 4938.68 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= \text{Rmom} / \text{QO} \\ &= 4938.68 / 6150.36 \\ &= 802.97 \text{ mm.} > \text{Bplen}/6 \rightarrow \text{Uplift in Transverse direction} \end{aligned}$$

$$\begin{aligned} f &= \text{Bplen} / 2 - \text{Edgedis} \\ &= 1200.0 / 2 - 50.8 \\ &= 549.20 \text{ mm.} \end{aligned}$$

Modular Ratio Of Steel/Concrete (n1):

$$\begin{aligned} &= \text{ES} / \text{EC} \\ &= 203402.5 / 21526.32 \\ &= 9.45 \end{aligned}$$

$$\begin{aligned} K1 &= 3 (e - 0.5 \cdot \text{Bplen}) \\ &= 3 (802.97 - 0.5 \cdot 1200.0) \\ &= 608.92 \text{ mm.} \end{aligned}$$

$$\begin{aligned} K2 &= 6 \cdot n1 \cdot \text{At} / \text{Bpwid} \cdot (f + e) \\ &= 6 \cdot 9.45 \cdot 4.87 / 225.0 \cdot (549.2 + 802.97) \\ &= 165838.74 \text{ mm.}^2 \end{aligned}$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot \text{Bplen} + f) \\ &= -165838.73 \cdot (0.5 \cdot 1200.0 + 549.2) \\ &= -190581868.92 \text{ mm.}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + 608.92 \cdot Y^2 + 165838.73 \cdot Y + -0.2E+09 &= 0 \\ Y &= 365.30 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Num} &= (\text{Bplen} / 2 - Y / 3 - e) \\ &= (1200.0 / 2 - 365.3 / 3 - 802.97) \\ &= -324.74 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (1200.0 / 2 - 365.3 / 3 + 549.2) \\ &= 1027.43 \end{aligned}$$

Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= - \text{QO} \cdot \text{Num} / \text{Denom} \\ &= - 6150.36 \cdot -324.74 / 1027.43 \\ &= 1943.93 \text{ Kg} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 1943.93 / (172.38 \cdot 1.0) \\ &= 1.1060 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 2.3589, 1.106] \\ &= 2.3589 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(\text{QO} + \text{Tforce}) / (Y \cdot \text{Bpwid})$$

$$= 2(6150.36 + 1943.93) / (365.3 \cdot 225.0)$$
$$= 19.31 \text{ bars}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (Bplen - WebLength) / 2$$
$$= (1200.0 - 1149.2) / 2$$
$$= 25.4000 \text{ mm.}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= Tforce \cdot Adist$$
$$= 1943.93 \cdot 25.4$$
$$= 49.38 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc \cdot (Y - Adist) / Y$$
$$= 19.31 \cdot (365.3 - 25.4) / 365.3$$
$$= 17.97 \text{ bars}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$
$$= (25.4^2 \cdot 225.0 / 6) \cdot (17.97 + 2 \cdot 19.31)$$
$$= 13.96 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$
$$= \sqrt{(6 \cdot \max(49.38, 13.96) / (225.0 \cdot 222.0))}$$
$$= 7.6267 \text{ mm.}$$

Review notes about nozzle loadings and supports in the Warnings report.

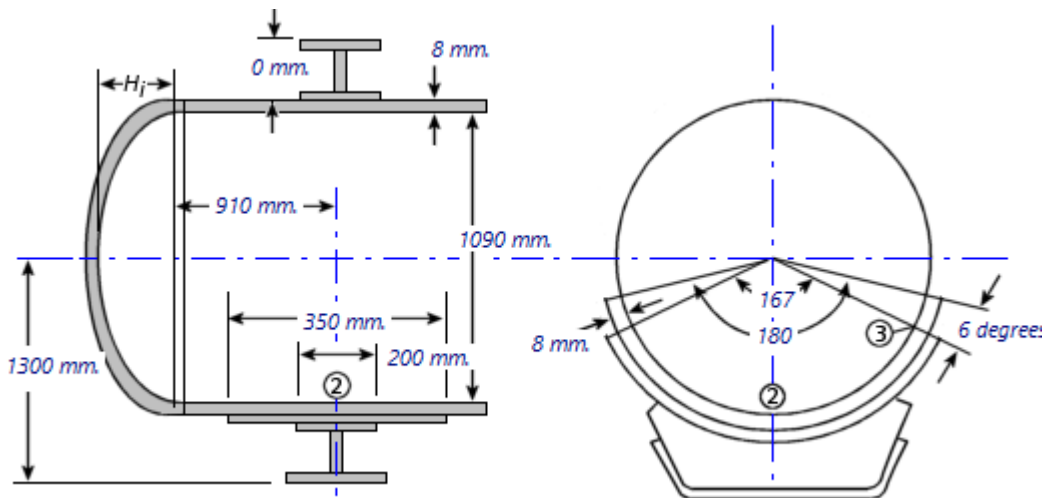
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	549.00	mm.
Shell Thickness used in this Case	t	8.000	mm.
Stiffened Vessel Length per 4.15.6	L	4600.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	910.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	350.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	8.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	613.00	mm.
Shell Allowable Stress used in Calculation		194.75	N./mm ²
Head Allowable Stress used in Calculation		194.75	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		7663.46	Kgf
Pressure used in Analysis	P	17.804	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		60.50	194.75
Long. Stress at Bottom of Midspan		61.69	194.75
Long. Stress at Top of Saddles		64.04	194.75
Long. Stress at Bottom of Saddles		58.15	194.75
Tangential Shear in Shell		2.80	109.26
Circ. Compressive Stress in Shell		1.10	194.75
Stiffener Circ. Stress at Shell		2.71	194.75
Stiffener Circ. Stress at Tip		15.25	194.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (866.0/2 + 0) \cdot 1300.0/1114.7876$
 $= 1514.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(81.28, 0) \cdot 1300.0/2790.0$
 $= 37.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 6149 + \max(38, 1515, 0, 0)$
 $= 7663.5 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.106 \cdot 1.2)^2)$
 $= 2415.358 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8085.14	Kgf
Transverse Shear Load	433.02	Kgf
Longitudinal Shear Load	81.28	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1090.0/2 + 8.0 + 8.0) \sin(167.0/2), 2 \cdot 549.0)$
 $= 1114.788 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1974	K2 = 0.6609	K3 = 0.3428	K4 = 0.2477
K5 = 0.6419	K6 = 0.0229	K7 = 0.0229	K8 = 0.2738
K9 = 0.1786	K10 = 0.0228	K1* = 0.3348	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -7663 \cdot 910.0 [1 - (1 - 910.0/4600.8 + (549.0^2 - 613.0^2) / (2 \cdot 910.0 \cdot 4600.8)) / (1 + (4 \cdot 613.0)/(3 \cdot 4600.8))]$
 $= -2275.9 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 7663 \cdot 4601/4 ((1 + 2(549^2 - 613^2)/(4601^2)) / (1 + (4 \cdot 613)/(3 \cdot 4601)) - 4 \cdot 910/4601)$
 $= 458.5 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 17.804 \cdot 549.0/(2 \cdot 8.0) - 458.5/(\pi \cdot 549.0^2 \cdot 8.0)$
 $= 60.50 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 17.804 \cdot 549.0/(2 \cdot 8.0) + 458.5/(\pi \cdot 549.0^2 \cdot 8.0)$
 $= 61.69 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
 $= 17.804 \cdot 549.0/(2 \cdot 8.0) - 2275.9/(\pi \cdot 549.0^2 \cdot 8.0)$
 $= 64.04 \text{ N./mm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 17.804 \cdot 549.0 / (2 \cdot 8.0) + -2275.9 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 58.15 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 7663(4600.8 - 2 \cdot 910.0) / (4600.8 + (4 \cdot 613.0 / 3))$$

$$= 3933.2 \text{ Kg}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 3933.19 / (\pi \cdot 549.0 \cdot 8.0)$$

$$= 2.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(549.0 \cdot 8.0)}$$

$$= 51.692 \text{ mm.}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.642 \cdot 7663 \cdot 0.1/43.85$$

$$= -1.10 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.2738 \cdot 7663/43.85 + 0.0229 \cdot 7663 \cdot 549.0 \cdot 48.87/622.72$$

$$= 2.71 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.2738 \cdot 7663/43.85 - 0.0229 \cdot 7663 \cdot 549.0 \cdot 69.73/622.72$$

$$= -15.25 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	30.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw,c	570.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(545.0 \cdot 8.0)} + 350.0) 8.0^3) / 12$$

$$= 1.933 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 1.933 + 36.241(376.426 - 4.0)^2$$

$$= 50268.043 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	453.0	8.0	4.0	36.2	14496.2	0.503E+05
Wearplate	350.0	8.0	12.0	28.0	33600.0	0.372E+05
Web	20.0	709.0	370.5	141.8	5253689.5	0.594E+05
BasePlate	225.0	30.0	740.0	67.5	4995000.0	0.893E+05
Totals	...	...	...	273.5	10296785.0	0.236E+06

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 10296785/273.541 \\ &= 376.426 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 167.0/2 \\ &= 96.5 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.684 + \sin(96.5) \cos(96.5)) \\ &= 0.2924 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.292 \cdot 7663.463 \\ &= 2240.5852 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 0.9260 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 186.0069 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1300.0 - 545.0 \cdot \sin(167.0/2) / 1.4574 \\ &= 928.437 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2080.2844 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 3.2514 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 206.6743 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(7663 + 422) \cdot 225.0 / (4 \cdot 1200.0 \cdot 206.674)} \\ &= 7.345 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 167.0/2) (545.0 + 8.0 + 8.0) \\ &= 1114.788 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 1114.7876 / (4 - 1) \\ &= 371.596 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 371.5959 \cdot 225.0/2 \\ &= 418.045 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 8085.135 / (1200.0 \cdot 225.0) \\ &= 2.994 \text{ Kg/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 418.0 \cdot 2.99 \\ &= 1251.835 \text{ Kg} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 36.0 + 37.16 \\ &= 73.160 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1251.8/73.1596$$

$$= 1.678 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B	D	Y	A	AY	Io
Rib+Web 20.0	200.0	...	40.0	...	0.133E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 20.0$$

$$= 180.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/73.16$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1333.3/73.16)}$$

$$= 42.691 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300.0 - \cos(167.0/2) (545.0 + 8.0 + 8.0) - 30.0$$

$$= 1206.493 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/310.0)}$$

$$= 112.834$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1206.493/42.691$$

$$= 28.261$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 81.3 / (2 \cdot 1200.0) \cdot 371.596 \cdot 1206.49/2$$

$$= 7.592 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($28.2612 < 112.8344$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (28.26)^2 / (2 \cdot 112.83^2)) 310 /$$

$$(5/3 + 3 \cdot (28.26) / (8 \cdot 112.83) - (28.26^3) / (8 \cdot 112.83^3))$$

$$= 170.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C_1 / I) / Sba$$

$$= 1.68/170.75 + (7.59 \cdot 100.0/13333333) / 206.67$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



Rib	20.0	180.0	0.0	36.0	0.0	0.133E+04
Web	371.6	20.0	0.0	74.3	0.0	24.8
Totals	...	...	...	110.3	...	0.136E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/110.319$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1300.0 - \sqrt{(561.0 + 8.0)^2 - (371.596/2)^2} - 30.0$$

$$= 740.661 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1356.8/110.319)}$$

$$= 35.069 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 740.661/35.069$$

$$= 21.120$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 81.28/(2 \cdot 1200.0)$$

$$= 0.034 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.034 \cdot 371.596 \cdot 740.661$$

$$= 9.321 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.321/135677.312$$

$$= 0.674 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (21.1199 < 112.8344) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.12)^2 / (2 \cdot 112.83^2)) 310 /$$

$$(5/3 + 3 \cdot (21.12) / (8 \cdot 112.83) - (21.12^3) / (8 \cdot 112.83^3))$$

$$= 175.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 2.23/175.45 + (9.32 \cdot 100.0/1356.772) / 206.67$$

$$= 0.016$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	207.01 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	6570.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	37.9 Kg
Maximum Transverse Force	Ft	433.0 Kg
Maximum Longitudinal Force	Fl	81.3 Kg

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $81.3 / (4.867 \cdot 2)$
= 0.8 N./mm^2 . Must be less than 103.2 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $433.0 / (4.867 \cdot 2)$
= 4.4 N./mm^2 . Must be less than 103.2 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $81.28 / (103.2 \cdot 2.0)$
= 0.0386 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1300.0 \cdot 433.02 + 0.0$
= 562.93 Kg-m .

Eccentricity (e):
= R_{mom} / QO
= $562.93 / 6570.26$
= 85.68 mm . < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.0386, 0.0]$
= 0.0386 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	549.00	mm.
Shell Thickness used in this Case	t	8.000	mm.
Stiffened Vessel Length per 4.15.6	L	4600.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	910.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	167.00	degrees
Wear Plate Width	b2 or b1	350.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	8.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	276.50	mm.
Shell Allowable Stress used in Calculation		194.75	N./mm ²
Head Allowable Stress used in Calculation		194.75	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 7659.98 Kgf
 Pressure used in Analysis P 17.804 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	59.33	194.75
Long. Stress at Bottom of Midspan	62.85	194.75
Long. Stress at Top of Saddles	63.19	194.75
Long. Stress at Bottom of Saddles	59.00	194.75
Tangential Shear in Shell	3.05	109.26
Circ. Compressive Stress in Shell	1.10	194.75
Stiffener Circ. Stress at Shell	2.71	194.75
Stiffener Circ. Stress at Tip	15.25	194.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(866.0/2 + 0) \cdot 1300.0/1114.7876$
 $= 1514.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(81.28, 0) \cdot 1300.0/2790.0$
 $= 37.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 6145 + \max(38, 1515, 0, 0)$
 $= 7660.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.106 \cdot 1.2)^2)$
 $= 2415.358 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 8081.66 Kgf
 Transverse Shear Load 433.02 Kgf
 Longitudinal Shear Load 81.28 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1090.0/2 + 8.0 + 8.0)\sin(167.0/2), 2 \cdot 549.0)$
 $= 1114.788 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1974 K2 = 0.6609 K3 = 0.3428 K4 = 0.2477
 K5 = 0.6419 K6 = 0.0229 K7 = 0.0229 K8 = 0.2738
 K9 = 0.1786 K10 = 0.0228 K1* = 0.3348 K6p = 0.0174
 K7p = 0.0174

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -7660 \cdot 910.0 [1 - (1 - 910.0/4600.8 + (549.0^2 - 276.5^2)/(2 \cdot 910.0 \cdot 4600.8))/(1 + (4 \cdot 276.5)/(3 \cdot 4600.8))]$
 $= -1620.2 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 7660 \cdot 4601 / 4 \cdot ((1 + 2(549^2 - 277^2) / (4601^2)) / (1 + (4 \cdot 277) / (3 \cdot 4601)) - 4 \cdot 910 / 4601)$$

$$= 1359.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 17.804 \cdot 549.0 / (2 \cdot 8.0) - 1359.7 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 59.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 17.804 \cdot 549.0 / (2 \cdot 8.0) + 1359.7 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 62.85 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= 17.804 \cdot 549.0 / (2 \cdot 8.0) - -1620.2 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 63.19 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= 17.804 \cdot 549.0 / (2 \cdot 8.0) + -1620.2 / (\pi \cdot 549.0^2 \cdot 8.0)$$

$$= 59.00 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7660(4600.8 - 2 \cdot 910.0) / (4600.8 + (4 \cdot 276.5 / 3))$$

$$= 4286.4 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot Rm \cdot t)$$

$$= 4286.35 / (\pi \cdot 549.0 \cdot 8.0)$$

$$= 3.05 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(549.0 \cdot 8.0)}$$

$$= 51.692 \text{ mm.}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K5 \cdot Q \cdot k / A$$

$$= -0.642 \cdot 7660 \cdot 0.1 / 43.85$$

$$= -1.10 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K8 \cdot Q / A + K6 \cdot Q \cdot Rm \cdot c1 / I$$

$$= -0.2738 \cdot 7660 / 43.85 + 0.0229 \cdot 7660 \cdot 549.0 \cdot 48.87 / 622.72$$

$$= 2.71 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K8 \cdot Q / A - K6 \cdot Q \cdot Rm \cdot c2 / I$$

$$= -0.2738 \cdot 7660 / 43.85 - 0.0229 \cdot 7660 \cdot 549.0 \cdot 69.73 / 622.72$$

$$= -15.25 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm.
Baseplate Thickness	Bpthk	30.0000	mm.
Baseplate Width	Bpwid	225.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	20.0000	mm.
Web Thickness	Webtk	20.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./mm ²
Height of Web at Center	Hw,c	570.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(545.0 \cdot 8.0)} + 350.0)^3) / 12$
 $= 1.933 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 1.933 + 36.241 (376.426 - 4.0)^2$
 $= 50268.043 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	453.0	8.0	4.0	36.2	14496.2	0.503E+05
Wearplate	350.0	8.0	12.0	28.0	33600.0	0.372E+05
Web	20.0	709.0	370.5	141.8	5253689.5	0.594E+05
BasePlate	225.0	30.0	740.0	67.5	4995000.0	0.893E+05
Totals	...	...	...	273.5	10296785.0	0.236E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 10296785 / 273.541$
 $= 376.426 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 167.0 / 2$
 $= 96.5$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(96.5) - 0.5 \cdot \sin(96.5)^2) / (\pi - 1.684 + \sin(96.5) \cos(96.5))$
 $= 0.2924$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.292 \cdot 7659.984$
 $= 2239.5681 \text{ Kg}$

Tension Stress, St = (Fh/As) = 0.9255 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 186.0069 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1300.0 - 545.0 \cdot \sin(167.0/2) / 1.4574$
 $= 928.437 \text{ mm.}$

Bending Moment, M = Fh * d = 2079.3398 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.2500 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 206.6743 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(7660 + 422)225.0 / (4 \cdot 1200.0 \cdot 206.674))}$
 $= 7.343 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 167.0/2) (545.0 + 8.0 + 8.0)$
 $= 1114.788 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nrifs} - 1)$
 $= 1114.7876 / (4 - 1)$
 $= 371.596 \text{ mm.}$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 371.5959 \cdot 225.0 / 2$$

$$= 418.045 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 8081.657 / (1200.0 \cdot 225.0)$$

$$= 2.993 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 418.0 \cdot 2.99$$

$$= 1251.296 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.0 + 37.16$$

$$= 73.160 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1251.3 / 73.1596$$

$$= 1.677 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	200.0	...	40.0	...	0.133E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 200.0 - 20.0$$

$$= 180.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 73.16$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1333.3 / 73.16) }$$

$$= 42.691 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1300.0 - \cos(167.0 / 2) (545.0 + 8.0 + 8.0) - 30.0$$

$$= 1206.493 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 310.0) }$$

$$= 112.834$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1206.493 / 42.691$$

$$= 28.261$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 81.3 / (2 \cdot 1200.0) \cdot 371.596 \cdot 1206.49 / 2$$

$$= 7.592 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (28.2612 < 112.8344) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (28.26)^2 / (2 \cdot 112.83^2)) 310 / (5/3 + 3 \cdot (28.26) / (8 \cdot 112.83) - (28.26^3) / (8 \cdot 112.83^3))$$

$$= 170.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.68 / 170.75 + (7.59 \cdot 100.0 / 13333333) / 206.67$$

$$= 0.013$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	180.0	0.0	36.0	0.0	0.133E+04
Web	371.6	20.0	0.0	74.3	0.0	24.8
Totals	...	...	...	110.3	...	0.136E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 110.319$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1300.0 - \sqrt{ (561.0 + 8.0)^2 - (371.596/2)^2 } - 30.0$$

$$= 740.661 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1356.8 / 110.319) }$$

$$= 35.069 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 740.661 / 35.069$$

$$= 21.120$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 81.28 / (2 \cdot 1200.0)$$

$$= 0.034 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.034 \cdot 371.596 \cdot 740.661$$

$$= 9.321 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.321 / 135677.312$$

$$= 0.674 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (21.1199 < 112.8344) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (21.12)^2 / (2 \cdot 112.83^2)) 310 / (5/3 + 3 \cdot (21.12) / (8 \cdot 112.83) - (21.12^3) / (8 \cdot 112.83^3))$$

$$= 175.4 \text{ N./mm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 2.23/175.45 + (9.32 \cdot 100.0/1356.772)/206.67$$

$$= 0.016$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	207.01 N./mm ²
Area Available in a Single Bolt	BlArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	6566.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	37.9 Kgf
Maximum Transverse Force	Ft	433.0 Kgf
Maximum Longitudinal Force	Fl	81.3 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 81.3 / (4.867 \cdot 2)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 433.0 / (4.867 \cdot 2)$$

$$= 4.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 81.28 / (103.2 \cdot 2.0)$$

$$= 0.0386 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1300.0 \cdot 433.02 + 0.0$$

$$= 562.93 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 562.93/6566.78$$

$$= 85.72 \text{ mm. < Bplen/6 --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0.0, 0.0386, 0.0]$$

$$= 0.0386 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M1-Man Hole

From: 10

Pressure for Reinforcement Calculations	P	191.002	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-240 304	+IT
Parent Allowable Stress at Temperature	Sv	129.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Hemispherical Head	D	1180.00	mm.
Head Finished (Minimum) Thickness	t	46.0000	mm.
Head Internal Corrosion Allowance	c	0.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

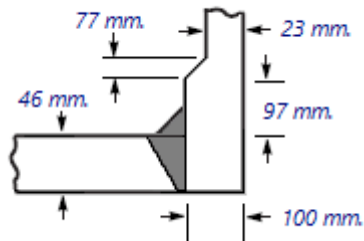
Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	110.00	N./mm ²
Allowable Stress At Ambient	Sna	117.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		200.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F304	
Hub Height of Integral Nozzle	h	96.9876	mm.
Height of Beveled Transition	L'	76.9876	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		100.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	48.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		2500	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M1-Man Hole

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	173.050	mm.
Actual Thickness Used in Calculation	23.012	mm.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 177.05 \cdot (\exp(191.002/(110.001 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 16.7875 + 2.0 + 0.0 = 18.7875 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 110.001 \cdot 1.0 \cdot \ln((2 \cdot 21.012 + 177.05) / 177.05) - 0.0$$

$$= 234.267 - 0.0 = 234.267 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.160 mm
Min. required shell thickness, internal p + c	45.360 mm.
Nozzle neck minimum required thickness	: 9.160 mm.

Minimum nozzle neck thickness 20.142 mm., is greater than required thickness 9.160 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[110.0/129.0, 1]$$

$$= 0.853$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 100.0 - 2.0$$

$$= 98.000 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 23.012 - 2.0$$

$$= 21.012 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1180.0/2 + 0.0$$

$$= 590.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{590.0 \cdot 46.0}, 2 \cdot 88.525)$$

$$= 164.742 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((46.0 + 0.0)/98.0)^{0.35}, 1)$$

$$= \min(0.767, 1)$$

$$= 0.767$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:
= $\min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t)$
= $\min(46.0 + 0.0 + 0.767 \cdot \sqrt{(88.525 \cdot 98.0)}, 150.0 + 46.0)$
= 117.480 mm.

Effective Vessel Thickness [teff]:
= $t + ((A_5 \cdot frp) / LR)$
= $46.0 + ((0.0 \cdot 1.0) / 164.742)$
= 46.000 mm.

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:
= $t \cdot LR$
= $46.0 \cdot 164.742$
= 75.781 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:
= $t_n \cdot LH$
= $98.0 \cdot 117.48$
= 115.130 cm²

Area Contributed by the Outside Fillet Weld [A41]:
= $0.5 \cdot Leg_1^2$
= $0.5 \cdot 10.0^2$
= 0.500 cm²

The total area contributed by A1 through A43 [AT]:
= $A_1 + frn(A_2 + A_3) + A_{41} + A_{42} + A_{43}$
= $75.781 + 0.853(115.13 + 0.0) + 0.5 + 0.0 + 0.0$
= 174.454 cm²

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $98.0 / \ln[1 + 98.0/88.525]$
= 131.494 mm.

Shell Radius for Force Calculation [Rxs]:
= $teff / \ln[1 + teff/Reff]$
= $46.0 / \ln[1 + 46.0/590.0]$
= 612.712 mm.

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S \cdot E$
= $1.5 \cdot 129.001 \cdot 1.0$
= 193.5 N./mm²

Force acting on the Nozzle [fN]:
= $P \cdot Rxn \cdot LH$
= $191.002 \cdot 131.494 \cdot 117.48$
= 30088.7 Kgf

Force acting on the Shell [fS]:
= $P \cdot Rxs \cdot (LR + t_n)/2$
= $191.002 \cdot 612.712 \cdot (164.742 + 98.0)/2$
= 156780.1 Kgf

Discontinuity Force from Internal Pressure [fY]:
= $(P \cdot Rxs \cdot R_{nc})/2$
= $(191.002 \cdot 612.712 \cdot 88.525)/2$
= 52823.5 Kgf

Area Resisting Internal Pressure [Ap]:
= $(fS + fY + fN)/P$
= $(156780.1 + 52823.5 + 30088.7)/191.002$
= 1230.6 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $Sallow / (2 \cdot Ap/AT - Rxs/(2 \cdot teff))$

$$= 193.502 / (2 \cdot 1230.608 / 174.454 - 612.712 / (2 \cdot 46.0))$$
$$= 259.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t/Rxs]$$
$$= 2 \cdot 129.001[46.0/612.712]$$
$$= 193.7 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(259.782, 193.686)$$
$$= 193.686 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (30088.727 + 156780.078 + 52823.531) / 174.454$$
$$= 134.741 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot Rxs / (2 \cdot teff)$$
$$= 191.002 \cdot 612.712 / (2 \cdot 46.0)$$
$$= 127.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 134.741 - 127.213, 127.213)$$
$$= 142.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	193.50	N./mm ²
Local Primary Membrane Stress	PL	142.27	N./mm ²
Maximum Allowable Working Pressure	Pmax	193.69	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (88.525 + 98.0) / 88.525$$
$$= 2.107 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (88.525 + 98.0)$$
$$= 292.993 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 98.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$
$$= \min(52824 \cdot 2.11, 1.5 \cdot 110.0(136.359 + 0.0), \pi/4 \cdot 191.0 \cdot 88.53^2 \cdot 2.11^2)$$
$$= 53222.941 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T))$$
$$= 53222.941 / (292.993(0.49 \cdot 7.071 + 0.6 \cdot 46.0 + 0.49 \cdot 0.0))$$
$$= 57.346 < \text{ or } = \text{ to } 129.001 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

The Drop for this Nozzle is : 30.2604 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2604 mm.

Input, Nozzle Desc: A1-Tube Inlet

From: 20

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-240 304	+IT
Parent Allowable Stress at Temperature	Sv	129.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1090.00	mm.
Shell Finished (Minimum) Thickness	t	135.0000	mm.
Shell Internal Corrosion Allowance	c	0.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		750.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

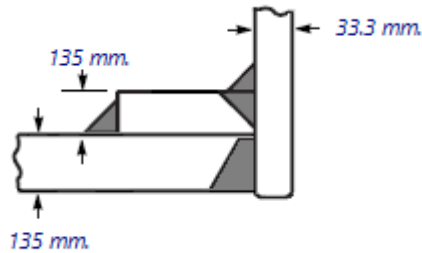
Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	129.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	135.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-240 304	+IT
Pad Allowable Stress at Temperature	Sp	129.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	423.8500	mm.
Thickness of Pad	te	135.0000	mm.
Weld leg size between Pad and Shell	Wp	120.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	100.0000	mm.
Reinforcing Pad Width		50.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		2500	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A1-Tube Inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	257.200	mm.
Actual Thickness Used in Calculation	33.325	mm.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 261.2 \cdot (\exp(191.0/(129.001 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 20.8429 + 2.0 + 0.0 = 22.8429 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 129.001 \cdot 1.0 \cdot \ln((2 \cdot 31.325 + 261.2) / 261.2) - 0.0$$

$$= 277.327 - 0.0 = 277.3266 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.340 mm
 Min. required shell thickness, internal p + c : 87.150 mm.

Nozzle neck minimum required thickness : 10.340 mm.

Minimum nozzle neck thickness 29.159 mm., is greater than required thickness 10.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.2,	Allowable	: 129.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 297.5 N./mm ²	Passed
Occasional	: 41.6,	Allowable	: 171.6 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 90.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[129.0/129.0, 1]$$

$$= 1.000$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[129.0/129.0, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 33.325 - 2.0$$

$$= 31.325 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1090.0/135.0$$

$$= 8.074 \text{ must be less than 400.}$$



Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1090.0/2 + 0.0$$

$$= 545.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(R_{eff} \cdot t) + W}$$

$$= \sqrt{(545.0 \cdot 135.0) + 50.0}$$

$$= 321.247 \text{ mm.}$$

Limit [LR2]:

$$= \sqrt{((R_{eff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((545.0 + 135.0) \cdot (135.0 + 135.0))}$$

$$= 428.486 \text{ mm.}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 130.6$$

$$= 261.200 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(321.247, 428.486, 261.2)$$

$$= 261.200 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 135.0, 135.0] + \sqrt{(130.6 \cdot 31.325)}$$

$$= 198.961 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 150.0$$

$$= 150.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(135.0 + 135.0)$$

$$= 2160.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[198.961, 150.0, 2160.0] + 135.0$$

$$= 285.000 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 135.0 + ((67.5 \cdot 1.0) / 261.2)$$

$$= 160.842 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 130.6 + 31.325) / (\sqrt{(1090.0 + 160.842) \cdot 160.842}))$$

$$= 0.652$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 135.0 \cdot 261.2 \cdot \max((0.652/5)^{0.85}, 1)$$

$$= 352.621 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 31.325 \cdot 285.0$$

$$= 89.276 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 120.0^2 \\ &= 72.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(50.0 \cdot 135.0, 261.2 \cdot 135.0) \\ &= 67.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 352.621 + 1.0(89.276 + 0.0) + 0.5 + 72.0 + 0.0 + 1.0(67.5) \\ &= 581.896 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 31.325 / \ln[1 + 31.325/130.6] \\ &= 145.702 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 160.842 / \ln[1 + 160.842/545.0] \\ &= 621.959 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 129.001 \cdot 1.0 \\ &= 193.5 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 191.0 \cdot 145.702 \cdot 285.0 \\ &= 80879.7 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 191.0 \cdot 621.959 (261.2 + 31.325) \\ &= 354368.2 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 191.0 \cdot 621.959 \cdot 130.6 \\ &= 158210.5 \text{ Kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (354368.2 + 158210.5 + 80879.7) / 191.0 \\ &= 3046.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 193.502 / (2 \cdot 3046.916 / 581.896 - 621.959 / 160.842) \\ &= 292.9 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 129.001[135.0/621.959] \\ &= 280.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(292.924, 279.988) \end{aligned}$$

= 279.988 bars

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (80879.727 + 354368.25 + 158210.516) / 581.896 \\ &= 100.017 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_s / t_{eff} \\ &= 191.0 \cdot 621.959 / 160.842 \\ &= 73.9 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 100.017 - 73.862, 73.862) \\ &= 126.2 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	193.50	N./mm ²
Local Primary Membrane Stress	PL	126.17	N./mm ²
Maximum Allowable Working Pressure	Pmax	279.99	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (130.6 + 31.325) / 130.6 \\ &= 1.240 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (130.6 + 31.325) \\ &= 254.351 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (130.6 + 31.325 + 50.0) \\ &= 332.891 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 84.852, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 7.071, \text{ must be } \geq \min(0.7 \cdot 31.325, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 84.852, \text{ must be } \geq \min(0.6 \cdot 135.000, 0.6 \cdot 135.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(158211 \cdot 1.24, 1.5 \cdot 129.0(89.276 + 0.0), \pi/4 \cdot 191.0 \cdot 130.6^2 \cdot 1.24^2) \\ &= 40109.531 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 40109.5 \cdot 135.0 \cdot 129 / (135.0 \cdot 129 + 135.0 \cdot 129) \\ &= 20054.766 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 40109.5 \cdot 135.0 \cdot 129 / (135.0 \cdot 129 + 135.0 \cdot 129) \\ &= 20054.766 \text{ Kg} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 20054.766 / (254.351 \cdot (0.6 \cdot 135.0 + 0.49 \cdot 0.0)) \\ &= 9.546 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T)) \\ &= 20054.766 / (254.351 \cdot (0.6 \cdot 100.0 + 0.49 \cdot 7.071)) \\ &= 12.184 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_3]:
 $= fwp / (L_t P \cdot (0.49 \cdot L_{42T}))$
 $= 20054.766 / (332.891 \cdot (0.49 \cdot 84.852))$
 $= 14.210 \text{ N./mm}^2$

Maximum Shear Stress in the Welds:
 $= \max(\tau_1, \tau_2, \tau_3)$
 $= \max(9.546, 12.184, 14.21)$
 $= 14.2$ must be less than or equal to 129.0 N./mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -10 °C

The Drop for this Nozzle is : 24.6104 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.6104 mm.

Input Echo, WRC297 Item 1, Description: A1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1090.000	mm.
Shell Thickness	135.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 129.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	257.200	mm.
Nozzle Thickness	33.3248	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 129.00	N./mm ²
Diameter of Reinforcing Pad	423.850	mm.
Thickness of Reinforcing Pad	135.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-13705.89	Kgf
Circumferential Shear	Vc	13705.89	Kgf
Longitudinal Shear	Vl	10279.39	Kgf
Circumferential Moment	Mc	10.69	Kg-m.
Longitudinal Moment	Ml	10.69	Kg-m.
Torsional Moment	Mt	12.33	Kg-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, D_v/T_v (10.074) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_v (2.399) is < 5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (9.718) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{680.0(135.0 - 0.0)}$
 $= 1323.701 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1360.000 mm.
 Nozzle Outside Diameter (d) = 323.850 mm.
 Vessel Thickness used (T) = 270.000 mm.
 Nozzle Thickness used (t) = 31.325 mm.
 $T / t = 8.619$
 $d / t = 10.338$

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.534$$

Nr/P = 0.183
 Mr/P = 0.112
 M0/P = 0.272
 N0/P = 0.267
 MrD/Mc = 0.111
 NrDL/Mc = 0.154
 M0d/Mc = 0.300
 N0DL/Mc = 0.124
 MrD/M1 = 0.111
 NrDL/M1 = 0.157
 M0D/M1 = 0.293
 N0DL/M1 = 0.035

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	50	30	50
Outplane Stress Summary	34	47	34	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	17	14	17	14

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Outplane Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress		34	47	34	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)		0	0	0	0
Outplane Bending (P)		2	-2	2	-2
Outplane Membrane (Mc)		0	0	0	0
Outplane Bending (Mc)		0	0	0	0
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		15	15	15	15
Outplane Stress Summary		18	12	18	12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)		0	0	0	0
Inplane Bending (P)		1	-1	1	-1
Inplane Membrane (Mc)		0	0	0	0
Inplane Bending (Mc)		0	0	0	0
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		30	50	30	50
Inplane Stress Summary		32	49	32	49

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		0	0	0	0
Shear Stress Summary		0	0	0	0

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		32	49	32	49

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	85	108	85	108
Hoop Stress Summary	85	108	85	108

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	42	42	42	42
Axial Stress Summary	64	30	64	30

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	88	109	88	109

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	85	108	85	108
Hoop Stress Summary	85	108	85	108

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	42	42	42	42



Axial Stress Summary 64 30 64 30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-6	-6	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	87	109	87	109

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(612.5(135.0 - 0.0))}$
 $= 1272.779 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1225.000 mm.
 Nozzle Outside Diameter (d) = 423.850 mm.
 Vessel Thickness used (T) = 135.000 mm.
 Nozzle Thickness used (t) = 81.325 mm.
 $T / t = 1.660$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.042$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.134
 Mr/P = 0.143
 M0/P = 0.065
 N0/P = 0.143
 MrD/Mc = 0.233
 NrDL/Mc = 0.176
 M0d/Mc = 0.135
 N0DL/Mc = 0.037
 MrD/Ml = 0.172
 NrDL/Ml = 0.136
 M0D/Ml = 0.100
 N0DL/Ml = 0.147

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	68	87	68	87
Outplane Stress Summary	72	85	72	85

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	41	28	41	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	72	85	72	85

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	38	32	38	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	87	68	87
Inplane Stress Summary	75	82	75	82

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	75	82	75	82

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	50	30	50	30	50	30	50
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	1	-1	1	-1
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	1	-1	1	-1	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	30.0	50.0	30.0	50.0	30.0	50.0	30.0	50.0
Pm+Pl (SUS)	30.0	50.0	30.0	50.0	30.0	50.0	30.0	50.0
Pm+Pl+Q (Total)	32.0	48.0	32.0	48.0	31.0	49.0	31.0	49.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	129.00	Passed
Pm+Pl (SUS)	50.00	193.50	Passed
Pm+Pl+Q (TOTAL)	49.00	387.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	87	68	87	68	87	68	87
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	6	-6	6	-6
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	6	-6	6	-6	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	68.0	87.0	68.0	87.0	68.0	87.0	68.0	87.0
Pm+Pl (SUS)	69.0	88.0	69.0	88.0	68.0	87.0	68.0	87.0
Pm+Pl+Q (Total)	71.0	86.0	71.0	86.0	74.0	81.0	74.0	81.0



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	87.00	129.00	Passed
Pm+Pl (SUS)	88.02	193.50	Passed
Pm+Pl+Q (TOTAL)	86.02	387.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	85	108	85	108	85	108	85	108
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	17	-17	17	-17	17	-17	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-6	-6	6	6
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	85.0	108.0	85.0	108.0	85.0	108.0	85.0	108.0
Pm+Pl (SUS)	86.6	109.0	86.6	109.0	85.9	108.6	85.9	108.6
Pm+Pl+Q (Total)	87.6	108.8	87.6	108.8	86.5	108.5	86.5	108.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	129.00	Passed
Pm+Pl (SUS)	109.02	193.50	Passed
Pm+Pl+Q (TOTAL)	108.80	387.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: B1-Tube Outlet
From: 20

Pressure for Reinforcement Calculations	P	191.004	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-240 304	+IT
Parent Allowable Stress at Temperature	Sv	129.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1090.00	mm.
Shell Finished (Minimum) Thickness	t	135.0000	mm.
Shell Internal Corrosion Allowance	c	0.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		750.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

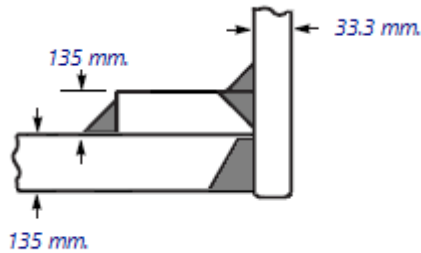
Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	129.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	135.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-240 304	+IT
Pad Allowable Stress at Temperature	Sp	129.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	423.8500	mm.
Thickness of Pad	te	135.0000	mm.
Weld leg size between Pad and Shell	Wp	120.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	100.0000	mm.
Reinforcing Pad Width		50.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		2500	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B1-Tube Outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	257.200	mm.
Actual Thickness Used in Calculation	33.325	mm.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 261.2 \cdot (\exp(191.004/(129.001 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 20.8434 + 2.0 + 0.0 = 22.8434 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 129.001 \cdot 1.0 \cdot \ln((2 \cdot 31.325 + 261.2) / 261.2) - 0.0$$

$$= 277.327 - 0.0 = 277.3266 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 152.19 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 423.13 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 10.340 mm
 Min. required shell thickness, internal p + c : 87.152 mm.

Nozzle neck minimum required thickness : 10.340 mm.

Minimum nozzle neck thickness 29.159 mm., is greater than required thickness 10.340 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 36.2,	Allowable	: 129.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 297.5 N./mm ²	Passed
Occasional	: 41.6,	Allowable	: 171.6 N./mm ²	Passed
Shear	: 15.1,	Allowable	: 90.3 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[129.0/129.0, 1]$$

$$= 1.000$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[129.0/129.0, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 33.325 - 2.0$$

$$= 31.325 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1090.0/135.0$$

$$= 8.074 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1090.0/2 + 0.0$$

$$= 545.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(545.0 \cdot 135.0) + 50.0}$$

$$= 321.247 \text{ mm.}$$

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + t_e))}$$

$$= \sqrt{((545.0 + 135.0) \cdot (135.0 + 135.0))}$$

$$= 428.486 \text{ mm.}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 130.6$$

$$= 261.200 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(321.247, 428.486, 261.2)$$

$$= 261.200 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 135.0, 135.0] + \sqrt{(130.6 \cdot 31.325)}$$

$$= 198.961 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 150.0$$

$$= 150.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(135.0 + 135.0)$$

$$= 2160.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[198.961, 150.0, 2160.0] + 135.0$$

$$= 285.000 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 135.0 + ((67.5 \cdot 1.0) / 261.2)$$

$$= 160.842 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 130.6 + 31.325) / (\sqrt{(1090.0 + 160.842) \cdot 160.842}))$$

$$= 0.652$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 135.0 \cdot 261.2 \cdot \max((0.652/5)^{0.85}, 1)$$

$$= 352.621 \text{ cm}^2$$



Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 31.325 \cdot 285.0 \\ &= 89.276 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 120.0^2 \\ &= 72.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(50.0 \cdot 135.0, 261.2 \cdot 135.0) \\ &= 67.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 352.621 + 1.0(89.276 + 0.0) + 0.5 + 72.0 + 0.0 + 1.0(67.5) \\ &= 581.896 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 31.325 / \ln[1 + 31.325/130.6] \\ &= 145.702 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 160.842 / \ln[1 + 160.842/545.0] \\ &= 621.959 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 129.001 \cdot 1.0 \\ &= 193.5 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 191.004 \cdot 145.702 \cdot 285.0 \\ &= 80881.5 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 191.004 \cdot 621.959 (261.2 + 31.325) \\ &= 354376.2 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 191.004 \cdot 621.959 \cdot 130.6 \\ &= 158214.1 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (354376.2 + 158214.1 + 80881.5) / 191.004 \\ &= 3046.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 193.502 / (2 \cdot 3046.916 / 581.896 - 621.959 / 160.842) \\ &= 292.9 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 129.001 [135.0 / 621.959] \\ &= 280.0 \text{ bars} \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(292.924, 279.988)
 = 279.988 bars

Average Primary Membrane Stress [σ_{Avg}]:
 = (fN + fS + fY) / AT
 = (80881.531 + 354376.188 + 158214.062) / 581.896
 = 100.019 N./mm²

General Primary Membrane Stress [σ_{Circ}]:
 = P • Rxs / teff
 = 191.004 • 621.959 / 160.842
 = 73.9 N./mm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
 = max(2 • 100.019 - 73.863, 73.863)
 = 126.2 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	193.50	N./mm ²
Local Primary Membrane Stress	PL	126.17	N./mm ²
Maximum Allowable Working Pressure	Pmax	279.99	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (130.6 + 31.325) / 130.6
 = 1.240 For set-in Nozzles

Weld Length of Nozzle to Shell Weld [LT]:
 = $\pi/2 \cdot (Rn + tn)$
 = $\pi/2 \cdot (130.6 + 31.325)$
 = 254.351 mm.

Weld Length of Pad to Shell Weld [LTp]:
 = $\pi/2 \cdot (Rn + tn + W)$
 = $\pi/2 \cdot (130.6 + 31.325 + 50.0)$
 = 332.891 mm.

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 84.852, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 31.325, 6.000)$ Passed
 tf1, L42T = 84.852, must be $\geq \min(0.6 \cdot 135.000, 0.6 \cdot 135.000)$ Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
 = min(158214 • 1.24, 1.5 • 129.0 (89.276 + 0.0), $\pi/4 \cdot 191.0 \cdot 130.6^2 \cdot 1.24^2$)
 = 40110.434 Kgf

Discontinuity Force [fws]:
 = fwelds • t • S / (t • S + te • Sp)
 = 40110.4 • 135.0 • 129 / (135.0 • 129 + 135.0 • 129)
 = 20055.217 Kgf

Discontinuity Force [fwp]:
 = fwelds • te • Sp / (t • S + te • Sp)
 = 40110.4 • 135.0 • 129 / (135.0 • 129 + 135.0 • 129)
 = 20055.217 Kgf

Shear Stress [τ_1]:
 = fws / (LT • (0.6 • tw1 + 0.49 • L43T))
 = 20055.217 / (254.351 • (0.6 • 135.0 + 0.49 • 0.0))
 = 9.546 N./mm²

Shear Stress [τ_2]:
 $= fwp / (L_T \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$
 $= 20055.217 / (254.351 \cdot (0.6 \cdot 100.0 + 0.49 \cdot 7.071))$
 $= 12.184 \text{ N./mm}^2$

Shear Stress [τ_3]:
 $= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$
 $= 20055.217 / (332.891 \cdot (0.49 \cdot 84.852))$
 $= 14.210 \text{ N./mm}^2$

Maximum Shear Stress in the Welds:
 $= \max(\tau_1, \tau_2, \tau_3)$
 $= \max(9.546, 12.184, 14.21)$
 $= 14.2 \text{ must be less than or equal to } 129.0 \text{ N./mm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -10 °C

Flange MDMT including temperature reduction per 3.11:
 Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 24.6104 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.6104 mm.

Input Echo, WRC297 Item 1, Description: B1-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1090.000	mm.
Shell Thickness	135.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 129.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	257.200	mm.
Nozzle Thickness	33.3248	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 129.00	N./mm ²
Diameter of Reinforcing Pad	423.850	mm.
Thickness of Reinforcing Pad	135.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure Dp 191.00 bars
 Radial Load P -13705.89 KgF

Circumferential Shear	Vc	13705.89	Kgf
Longitudinal Shear	Vl	10279.39	Kgf
Circumferential Moment	Mc	10.69	Kg-m.
Longitudinal Moment	Ml	10.69	Kg-m.
Torsional Moment	Mt	12.33	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, D_v/T_v (10.074) is < 20
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_v (2.399) is < 5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, D_n/T_n (9.718) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(680.0(135.0 - 0.0))}$
 $= 1323.701 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1360.000 mm.
Nozzle Outside Diameter	(d) =	323.850 mm.
Vessel Thickness used	(T) =	270.000 mm.
Nozzle Thickness used	(t) =	31.325 mm.
T / t	=	8.619
d / t	=	10.338

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.534$$

Nr/P	=	0.183
Mr/P	=	0.112
M0/P	=	0.272
N0/P	=	0.267
MrD/Mc	=	0.111
NrDL/Mc	=	0.154
M0d/Mc	=	0.300
N0DL/Mc	=	0.124
MrD/Ml	=	0.111
NrDL/Ml	=	0.157
M0D/Ml	=	0.293
N0DL/Ml	=	0.035

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	30	50	30	50
Outplane Stress Summary	34	47	34	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1



Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	15	15	15	15

Inplane Stress Summary	17	14	17	14
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	34	47	34	47
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15

Outplane Stress Summary	18	12	18	12
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	50	30	50

Inplane Stress Summary	32	49	32	49
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	0	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities)	Left	Left	Right	Right

	Outside	Inside	Outside	Inside
Two * Max Shear Stress	32	49	32	49

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	85	108	85	108
Hoop Stress Summary	85	108	85	108

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	42	42	42	42
Axial Stress Summary	64	30	64	30

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	8	8	-8	-8
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	88	109	88	109

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	85	108	85	108
Hoop Stress Summary	85	108	85	108

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	4	4	4	4
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	42	42	42	42

Axial Stress Summary	64	30	64	30
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-6	-6	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	87	109	87	109

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(612.5(135.0 - 0.0))}$
 $= 1272.779 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	1225.000 mm.
Nozzle Outside Diameter (d)	=	423.850 mm.
Vessel Thickness used (T)	=	135.000 mm.
Nozzle Thickness used (t)	=	81.325 mm.
T / t	=	1.660
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.042$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.134
Mr/P	=	0.143
M0/P	=	0.065
N0/P	=	0.143
MrD/Mc	=	0.233
NrDL/Mc	=	0.176
M0d/Mc	=	0.135
N0DL/Mc	=	0.037
MrD/Ml	=	0.172
NrDL/Ml	=	0.136
M0D/Ml	=	0.100
N0DL/Ml	=	0.147

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	68	87	68	87

Outplane Stress Summary	72	85	72	85
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	34	34	34

Inplane Stress Summary	41	28	41	28
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	1	1	-1	-1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	72	85	72	85
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34

Outplane Stress Summary	38	32	38	32
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	87	68	87

Inplane Stress Summary	75	82	75	82
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	75	82	75	82

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	50	30	50	30	50	30	50
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	1	-1	1	-1
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	1	-1	1	-1	2	-2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	30.0	50.0	30.0	50.0	30.0	50.0	30.0	50.0
Pm+Pl (SUS)	30.0	50.0	30.0	50.0	30.0	50.0	30.0	50.0
Pm+Pl+Q (Total)	32.0	48.0	32.0	48.0	31.0	49.0	31.0	49.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	129.00	Passed
Pm+Pl (SUS)	50.00	193.50	Passed
Pm+Pl+Q (TOTAL)	49.00	387.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	87	68	87	68	87	68	87
Circ. Pl (SUS)	1	1	1	1	0	0	0	0
Circ. Q (SUS)	2	-2	2	-2	6	-6	6	-6
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	0	0	0	0	1	1	1	1
Long. Q (SUS)	6	-6	6	-6	2	-2	2	-2

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	68.0	87.0	68.0	87.0	68.0	87.0	68.0	87.0
Pm+P1 (SUS)	69.0	88.0	69.0	88.0	68.0	87.0	68.0	87.0
Pm+P1+Q (Total)	71.0	86.0	71.0	86.0	74.0	81.0	74.0	81.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	87.00	129.00	Passed
Pm+P1 (SUS)	88.02	193.50	Passed
Pm+P1+Q (TOTAL)	86.02	387.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	85	108	85	108	85	108	85	108
Circ. P1 (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. P1 (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	17	-17	17	-17	17	-17	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	8	8	-8	-8	-6	-6	6	6
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	85.0	108.0	85.0	108.0	85.0	108.0	85.0	108.0
Pm+P1 (SUS)	86.6	109.0	86.6	109.0	85.9	108.6	85.9	108.6
Pm+P1+Q (Total)	87.6	108.8	87.6	108.8	86.5	108.5	86.5	108.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	129.00	Passed
Pm+P1 (SUS)	109.02	193.50	Passed
Pm+P1+Q (TOTAL)	108.80	387.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: A2-Shell inlet From: 30

Pressure for Reinforcement Calculations	P	10.000	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1090.00	mm.
Shell Finished (Minimum) Thickness	t	8.0000	mm.
Shell Internal Corrosion Allowance	c	0.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2060.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

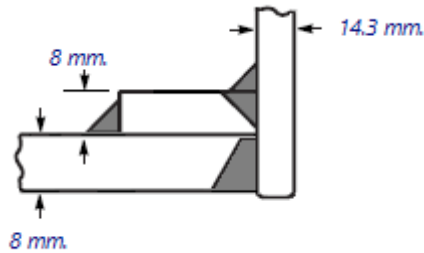
Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	146.00	N./mm ²
Allowable Stress At Ambient	Sna	165.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm.
Thickness of Pad	te	8.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm.
Reinforcing Pad Width		50.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A2-Shell inlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	139.725	mm.
Actual Thickness Used in Calculation	14.275	mm.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 143.725 \cdot (\exp(10.0/(145.998 \cdot 1.0)) - 1) + 2.0 + 0.0$
 $= 0.4939 + 2.0 + 0.0 = 2.4939 \text{ mm. Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 145.998 \cdot 1.0 \cdot \ln((2 \cdot 12.275 + 143.725) / 143.725) - 0.0$
 $= 230.219 - 0.0 = 230.219 \text{ bars}$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.220 mm
Min. required shell thickness, internal p + c :	3.964 mm.
Nozzle neck minimum required thickness :	3.964 mm.

Minimum nozzle neck thickness 12.497 mm., is greater than required thickness 3.964 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	1.4,	Allowable :	146.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	387.3 N./mm ²	Passed
Occasional :	3.2,	Allowable :	194.2 N./mm ²	Passed
Shear :	4.8,	Allowable :	102.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[146.0/138.0, 1]$
 $= 1.000$

Pad Material Factor [frp]:
 $= \min[S_p/S, 1]$
 $= \min[138.0/138.0, 1]$
 $= 1.000$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 14.275 - 2.0$
 $= 12.275 \text{ mm.}$

Shell Diameter to Thickness ratio [D/t]:
 $= D/t$
 $= 1090.0/8.0$
 $= 136.250 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$



$$= 1090.0/2 + 0.0$$

$$= 545.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(R_{eff} \cdot t) + W}$$

$$= \sqrt{(545.0 \cdot 8.0) + 50.0}$$

$$= 116.030 \text{ mm.}$$

Limit [LR2]:

$$= \sqrt{((R_{eff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((545.0 + 8.0) \cdot (8.0 + 8.0))}$$

$$= 94.064 \text{ mm.}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 71.863$$

$$= 143.725 \text{ mm.}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(116.03, 94.064, 143.725)$$

$$= 94.064 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 8.0, 8.0] + \sqrt{(71.863 \cdot 12.275)}$$

$$= 37.700 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(8.0 + 8.0)$$

$$= 128.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[37.7, 200.0, 128.0] + 8.0$$

$$= 45.700 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 8.0 + ((4.0 \cdot 1.0) / 94.064)$$

$$= 12.252 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 71.86 + 12.275) / (\sqrt{(1090.0 + 12.252) \cdot 12.252}))$$

$$= 1.342$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 8.0 \cdot 94.064 \cdot \max((1.342/5)^{0.85}, 1)$$

$$= 7.525 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 12.275 \cdot 45.7$$

$$= 5.610 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 8.0^2 \\ &= 0.320 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(50.0 \cdot 8.0, 94.064 \cdot 8.0) \\ &= 4.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 7.525 + 1.0(5.61 + 0.0) + 0.5 + 0.32 + 0.0 + 1.0(4.0) \\ &= 17.955 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 12.275 / \ln[1 + 12.275/71.863] \\ &= 77.839 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 12.252 / \ln[1 + 12.252/545.0] \\ &= 551.103 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 10.0 \cdot 77.839 \cdot 45.7 \\ &= 362.8 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs}(LR + t_n) \\ &= 10.0 \cdot 551.103(94.064 + 12.275) \\ &= 5976.1 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 10.0 \cdot 551.103 \cdot 71.863 \\ &= 4038.6 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (5976.1 + 4038.6 + 362.8) / 10.0 \\ &= 1017.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 206.996 / (2 \cdot 1017.645 / 17.955 - 551.103 / 12.252) \\ &= 30.3 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 / [8.0 / 551.103] \\ &= 20.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(30.271, 20.031) \\ &= 20.031 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (362.753 + 5976.131 + 4038.617) / 17.955 \\ &= 56.682 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 10.0 \cdot 551.103 / 12.252 \\ &= 45.0 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 56.682 - 44.982, 44.982) \\ &= 68.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	68.38	N./mm ²
Maximum Allowable Working Pressure	Pmax	20.03	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (71.863 + 12.275) / 71.863 \\ &= 1.171 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (71.863 + 12.275) \\ &= 132.163 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (71.863 + 12.275 + 50.0) \\ &= 210.703 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 5.657, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 12.275, 6.000)$ Passed

tf1, L42T = 5.657, must be $\geq \min(0.6 \cdot 8.000, 0.6 \cdot 8.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(4039 \cdot 1.17, 1.5 \cdot 146.0(5.61 + 0.0), \pi/4 \cdot 10.0 \cdot 71.86^2 \cdot 1.17^2) \\ &= 566.977 \text{ Kg} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 567.0 \cdot 8.0 \cdot 138 / (8.0 \cdot 138 + 8.0 \cdot 138) \\ &= 283.489 \text{ Kg} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 567.0 \cdot 8.0 \cdot 138 / (8.0 \cdot 138 + 8.0 \cdot 138) \\ &= 283.489 \text{ Kg} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 283.489 / (132.163 \cdot (0.6 \cdot 8.0 + 0.49 \cdot 0.0)) \\ &= 4.382 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 283.489 / (132.163 \cdot (0.6 \cdot 8.0 + 0.49 \cdot 7.071)) \\ &= 2.545 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$= fwp / (LTp \cdot (0.49 \cdot L42T))$$

$$= 283.489 / (210.703 \cdot (0.49 \cdot 5.657))$$
$$= 4.760 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$
$$= \max(4.382, 2.545, 4.76)$$
$$= 4.8 \text{ must be less than or equal to } 138.0 \text{ N./mm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.042

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (0.437 \cdot 1.0) / (12.49 - 2.0 - 0.0)$
 $= 0.042$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -21 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
Temperature reduction per Graph 3.12 : 20 °C
The computed stress ratio is : 0.495

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.964 \cdot 1.0) / (8.0 - 0.0 - 0.0)$
 $= 0.495$

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : -21 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : B
From Figure 3.7M- Non PWHT, the MDMT is : -1 °C
Temperature reduction per Graph 3.12 : 20 °C
The computed stress ratio is : 0.495

The computed thickness ratio [sr]:
 $= (tr \cdot E^*) / (tnom - ci - cext)$
 $= (3.964 \cdot 1.0) / (8.0 - 0.0 - 0.0)$
 $= 0.495$

Gov. MDMT of the Nozzle : -21 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 6.5338 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.5338 mm.

Input Echo, WRC297 Item 1, Description: A2-Shell inlet:

Diameter Basis for Cylindrical Shell ID
Shell Internal Corrosion Allowance 0.0000 mm.



Shell External Corrosion Allowance		0.0000	mm.
Shell Diameter		1090.000	mm.
Shell Thickness		8.0000	mm.
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-240 304	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²
Diameter Basis for Nozzle		ID	
Nozzle Corrosion Allowance		2.0000	mm.
Nozzle Diameter		139.725	mm.
Nozzle Thickness		14.2748	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	165.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	146.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	268.275	mm.
Thickness of Reinforcing Pad	8.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	10.00	bars
Radial Load	P	-917.81	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	688.36	Kgf
Circumferential Moment	Mc	0.36	Kg-m.
Longitudinal Moment	Ml	0.36	Kg-m.
Torsional Moment	Mt	0.41	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(553.0(8.0 - 0.0))}$
 $= 387.768 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1106.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	16.000	mm.
Nozzle Thickness used	(t) =	12.275	mm.
T / t	=	1.303	
d / t	=	13.709	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.265$$

Nr/P	=	0.107
Mr/P	=	0.149
M0/P	=	0.059
N0/P	=	0.130
MrD/Mc	=	0.253
NrDL/Mc	=	0.144
M0d/Mc	=	0.116
N0DL/Mc	=	0.063
MrD/Ml	=	0.167
NrDL/Ml	=	0.105
M0D/Ml	=	0.077
N0DL/Ml	=	0.159

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	50	26	50	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	51	-10	52	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	53	37	53	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	33	8	33	8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3

Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34

Inplane Stress Summary	68	6	68	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	68	9	68	9
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	11	12	11	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	38	-29	39	-29
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	2	2	-2	-2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress 39 42 39 42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	11	12	11	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	38	-29	39	-29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-2	-2	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	39	41	39	42

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{549.0(8.0 - 0.0)}$
 $= 386.973 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1098.000 mm.
 Nozzle Outside Diameter (d) = 268.275 mm.
 Vessel Thickness used (T) = 8.000 mm.
 Nozzle Thickness used (t) = 62.275 mm.
 $T / t = 0.128$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.862$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.064
 Mr/P = 0.111
 MØ/P = 0.033
 NØ/P = 0.049
 MrD/Mc = 0.271
 NrDL/Mc = 0.106
 MØd/Mc = 0.081
 NØDL/Mc = 0.069
 MrD/Ml = 0.084
 NrDL/Ml = 0.082
 MØD/Ml = 0.025
 NØDL/Ml = 0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	67	68	67	68
Outplane Stress Summary	102	47	102	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	136	-50	136	-51

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	136	99	136	99

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0

Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	33	33	33
Outplane Stress Summary	68	13	68	12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	67	68	67	68
Inplane Stress Summary	170	-15	170	-16

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	170	29	170	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	4	4	4	4	3	3	3	3
Circ. Q (SUS)	12	-12	12	-12	31	-31	31	-31
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	3	3	3	3	4	4	4	4
Long. Q (SUS)	31	-31	31	-31	12	-12	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0
Pm+Pl (SUS)	37.2	38.2	37.2	38.2	36.1	37.1	36.1	37.1
Pm+Pl+Q (Total)	51.6	38.2	51.6	38.2	67.0	8.4	67.0	8.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
------------------------	-----------	----------------	--------

Pm (SUS)	34.00	138.00	Passed
Pm+Pl (SUS)	38.21	207.00	Passed
Pm+Pl+Q (TOTAL)	67.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	67	68	67	68	67	68	67	68
Circ. Pl (SUS)	6	6	6	6	8	8	8	8
Circ. Q (SUS)	27	-27	27	-27	93	-93	93	-93
Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	8	8	8	8	6	6	6	6
Long. Q (SUS)	93	-93	93	-93	27	-27	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	67.0	68.0	67.0	68.0	67.0	68.0	67.0	68.0
Pm+Pl (SUS)	73.1	74.1	73.1	74.1	75.0	76.0	75.0	76.0
Pm+Pl+Q (Total)	134.1	99.1	134.1	99.1	168.0	29.1	168.0	29.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	138.00	Passed
Pm+Pl (SUS)	76.03	207.00	Passed
Pm+Pl+Q (TOTAL)	168.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	4	4	4	4	4	4	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	34	-34	34	-34	34	-34	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	10.6	11.5	10.6	11.5	10.6	11.5	10.6	11.5
Pm+Pl+Q (Total)	38.1	41.2	38.1	41.2	38.1	41.2	38.1	41.2



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	146.00	Passed
Pm+Pl (SUS)	11.53	219.00	Passed
Pm+Pl+Q (TOTAL)	41.19	438.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: B2-Shell outlet

From: 30

Pressure for Reinforcement Calculations	P	10.107	bars
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1090.00	mm.
Shell Finished (Minimum) Thickness	t	8.0000	mm.
Shell Internal Corrosion Allowance	c	0.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2060.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

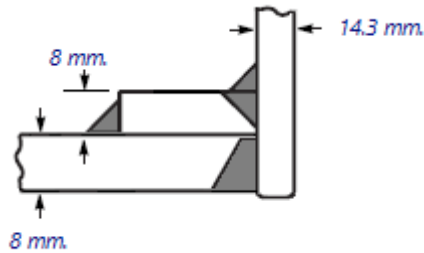
Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	2.0000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	270.0000	mm.
Thickness of Pad	te	8.0000	mm.
Weld leg size between Pad and Shell	Wp	8.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	8.0000	mm.
Reinforcing Pad Width		50.8625	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: B2-Shell outlet

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	139.725	mm.
Actual Thickness Used in Calculation	14.275	mm.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 143.725 \cdot (\exp(10.107/(137.997 \cdot 1.0)) - 1) + 2.0 + 0.0$$

$$= 0.5283 + 2.0 + 0.0 = 2.5283 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 137.997 \cdot 1.0 \cdot \ln((2 \cdot 12.275 + 143.725) / 143.725) - 0.0$$

$$= 217.603 - 0.0 = 217.6026 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	8.220 mm
Min. required shell thickness, internal p + c :	4.006 mm.
Nozzle neck minimum required thickness :	4.006 mm.

Minimum nozzle neck thickness 12.497 mm., is greater than required thickness 4.006 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	1.5,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	343.5 N./mm ²	Passed
Occasional :	3.2,	Allowable :	183.5 N./mm ²	Passed
Shear :	4.8,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/138.0, 1]$$

$$= 1.000$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[138.0/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 14.275 - 2.0$$

$$= 12.275 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$

$$= 1090.0/8.0$$

$$= 136.250 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$



$$= 1090.0/2 + 0.0$$

$$= 545.000 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(545.0 \cdot 8.0) + 50.863}$$

$$= 116.893 \text{ mm.}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((545.0 + 8.0) \cdot (8.0 + 8.0))}$$

$$= 94.064 \text{ mm.}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 71.863$$

$$= 143.725 \text{ mm.}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(116.893, 94.064, 143.725)$$

$$= 94.064 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 8.0, 8.0] + \sqrt{(71.863 \cdot 12.275)}$$

$$= 37.700 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 200.0$$

$$= 200.000 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(8.0 + 8.0)$$

$$= 128.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[\text{LH1}, \text{LH2}, \text{LH3}] + t \quad \text{for set-in nozzles}$$

$$= \min[37.7, 200.0, 128.0] + 8.0$$

$$= 45.700 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / \text{LR})$$

$$= 8.0 + ((4.069 \cdot 1.0) / 94.064)$$

$$= 12.326 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$

$$= \min(12, (2 \cdot 71.86 + 12.275) / (\sqrt{(1090.0 + 12.326) \cdot 12.326}))$$

$$= 1.338$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot \text{LR} \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 8.0 \cdot 94.064 \cdot \max((1.338/5)^{0.85}, 1)$$

$$= 7.525 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot \text{LH}$$

$$= 12.275 \cdot 45.7$$

$$= 5.610 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:



$$\begin{aligned} &= 0.5 \cdot \text{Leg42}^2 \\ &= 0.5 \cdot 8.0^2 \\ &= 0.320 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, LR \cdot t_e) \\ &= \min(50.863 \cdot 8.0, 94.064 \cdot 8.0) \\ &= 4.069 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 7.525 + 1.0(5.61 + 0.0) + 0.5 + 0.32 + 0.0 + 1.0(4.069) \\ &= 18.024 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 12.275 / \ln[1 + 12.275/71.863] \\ &= 77.839 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 12.326 / \ln[1 + 12.326/545.0] \\ &= 551.139 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 137.997 \cdot 1.0 \\ &= 207.0 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 10.107 \cdot 77.839 \cdot 45.7 \\ &= 366.6 \text{ Kg} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} (LR + t_n) \\ &= 10.107 \cdot 551.139 (94.064 + 12.275) \\ &= 6040.4 \text{ Kg} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 10.107 \cdot 551.139 \cdot 71.863 \\ &= 4082.0 \text{ Kg} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (6040.4 + 4082.0 + 366.6) / 10.107 \\ &= 1017.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 206.996 / (2 \cdot 1017.71 / 18.024 - 551.139 / 12.326) \\ &= 30.3 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 137.997 [8.0 / 551.139] \\ &= 20.0 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(30.343, 20.03) \\ &= 20.030 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (366.629 + 6040.382 + 4082.037) / 18.024 \\ &= 57.072 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 10.107 \cdot 551.139 / 12.326 \\ &= 45.2 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 57.072 - 45.195, 45.195) \\ &= 68.9 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	68.95	N./mm ²
Maximum Allowable Working Pressure	Pmax	20.03	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (71.863 + 12.275) / 71.863 \\ &= 1.171 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (71.863 + 12.275) \\ &= 132.163 \text{ mm.} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTP]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn + W) \\ &= \pi/2 \cdot (71.863 + 12.275 + 50.863) \\ &= 212.057 \text{ mm.} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 5.657, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 12.275, 6.000) \text{ Passed} \\ &tf1, L42T = 5.657, \text{ must be } \geq \min(0.6 \cdot 8.000, 0.6 \cdot 8.000) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(4082 \cdot 1.17, 1.5 \cdot 138.0(5.61 + 0.0), \pi/4 \cdot 10.1 \cdot 71.86^2 \cdot 1.17^2) \\ &= 573.035 \text{ Kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 573.0 \cdot 8.0 \cdot 138 / (8.0 \cdot 138 + 8.0 \cdot 138) \\ &= 286.517 \text{ Kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp) \\ &= 573.0 \cdot 8.0 \cdot 138 / (8.0 \cdot 138 + 8.0 \cdot 138) \\ &= 286.517 \text{ Kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (LT \cdot (0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 286.517 / (132.163 \cdot (0.6 \cdot 8.0 + 0.49 \cdot 0.0)) \\ &= 4.429 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (LT \cdot (0.6 \cdot tw2 + 0.49 \cdot L41T)) \\ &= 286.517 / (132.163 \cdot (0.6 \cdot 8.0 + 0.49 \cdot 7.071)) \\ &= 2.572 \text{ N./mm}^2 \end{aligned}$$

Shear Stress [τ_3]:

$$= fwp / (LTP \cdot (0.49 \cdot L42T))$$

$$= 286.517 / (212.057 \cdot (0.49 \cdot 5.657))$$

$$= 4.780 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(4.429, 2.572, 4.78)$$

$$= 4.8 \text{ must be less than or equal to } 138.0 \text{ N./mm}^2$$

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

The Drop for this Nozzle is : 6.5338 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.5338 mm.

Input Echo, WRC297 Item 1, Description: B2-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1090.000	mm.
Shell Thickness	8.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	139.725	mm.
Nozzle Thickness	14.2748	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²
Diameter of Reinforcing Pad	270.000	mm.
Thickness of Reinforcing Pad	8.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	10.11	bars
Radial Load	P	-917.81	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	688.36	Kgf
Circumferential Moment	Mc	0.36	Kg-m.
Longitudinal Moment	Ml	0.36	Kg-m.
Torsional Moment	Mt	0.41	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{553.0(8.0 - 0.0)}$$

$$= 387.768 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1106.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	16.000	mm.
Nozzle Thickness used	(t) =	12.275	mm.
T / t	=	1.303	

$$d / t = 13.709$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.265$$

Nr/P	=	0.107
Mr/P	=	0.149
M0/P	=	0.059
N0/P	=	0.130
MrD/Mc	=	0.253
NrDL/Mc	=	0.144
M0d/Mc	=	0.116
N0DL/Mc	=	0.063
MrD/Ml	=	0.167
NrDL/Ml	=	0.105
M0D/Ml	=	0.077
N0DL/Ml	=	0.159

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	50	27	50	27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	52	-10	52	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	53	37	53	38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	33	9	34	9
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34

Inplane Stress Summary	68	7	69	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	69	10	69	9
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	11	12	11	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary 38 -29 39 -29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 2 2 -2 -2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	39	42	39	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary 11 12 11 12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	34	-34	34	-34
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary 38 -29 39 -29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0

Shear Stress Summary -2 -2 2 2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	39	41	39	42

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(549.0(8.0 - 0.0))}$$

$$= 386.973 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1098.000 mm.
 Nozzle Outside Diameter (d) = 270.000 mm.
 Vessel Thickness used (T) = 8.000 mm.
 Nozzle Thickness used (t) = 63.137 mm.
 T / t = 0.127
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.881$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.064
 Mr/P = 0.111
 M0/P = 0.033
 N0/P = 0.048
 MrD/Mc = 0.271
 NrDL/Mc = 0.106
 M0d/Mc = 0.081
 N0DL/Mc = 0.069
 MrD/Ml = 0.083
 NrDL/Ml = 0.082
 M0D/Ml = 0.024
 N0DL/Ml = 0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	68	69	68	69
Outplane Stress Summary	102	48	102	48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	136	-50	136	-50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	136	99	136	99

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	68	13	68	13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	93	-93	93	-93
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	69	68	69
Inplane Stress Summary	170	-14	171	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	170	28	171	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	4	4	4	4	3	3	3	3
Circ. Q (SUS)	12	-12	12	-12	31	-31	31	-31
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	3	3	3	3	4	4	4	4

Long. Q (SUS)	31	-31	31	-31	12	-12	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0
Pm+P1 (SUS)	37.2	38.2	37.2	38.2	36.1	37.1	36.1	37.1
Pm+P1+Q (Total)	51.6	38.2	51.6	38.2	67.0	8.4	67.0	8.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	138.00	Passed
Pm+P1 (SUS)	38.21	207.00	Passed
Pm+P1+Q (TOTAL)	67.03	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	69	68	69	68	69	68	69
Circ. P1 (SUS)	6	6	6	6	8	8	8	8
Circ. Q (SUS)	27	-27	27	-27	93	-93	93	-93
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. P1 (SUS)	8	8	8	8	6	6	6	6
Long. Q (SUS)	93	-93	93	-93	27	-27	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	68.0	69.0	68.0	69.0	68.0	69.0	68.0	69.0
Pm+P1 (SUS)	74.1	75.1	74.1	75.1	76.0	77.0	76.0	77.0
Pm+P1+Q (Total)	135.1	99.1	135.1	99.1	169.0	29.1	169.0	29.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	69.00	138.00	Passed
Pm+P1 (SUS)	77.03	207.00	Passed
Pm+P1+Q (TOTAL)	169.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7



Circ. P1 (SUS)	4	4	4	4	4	4	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	34	-34	34	-34	34	-34	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+P1 (SUS)	10.6	11.5	10.6	11.5	10.6	11.5	10.6	11.5
Pm+P1+Q (Total)	38.1	41.2	38.1	41.2	38.1	41.2	38.1	41.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+P1 (SUS)	11.53	207.00	Passed
Pm+P1+Q (TOTAL)	41.19	413.99	Passed

*Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	10.00 bars
Shell Maximum Operating Pressure	Psox,max	10.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	8.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1090.000 mm.
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-240 304

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	191.00 bars
Channel Maximum Operating Pressure	Ptox,max	191.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	135.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1090.000 mm.
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	129.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1348
Tube Wall Thickness	et	2.1000 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	4300.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		4040.00 mm.
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		129.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1000 mm.
Groove Weld Leg	ag	2.1000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	512.600 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.
Area of Tube Layout	Ap	7543.9 cm ²
Perimeter of Tube Layout	Cp	3216.8 mm.

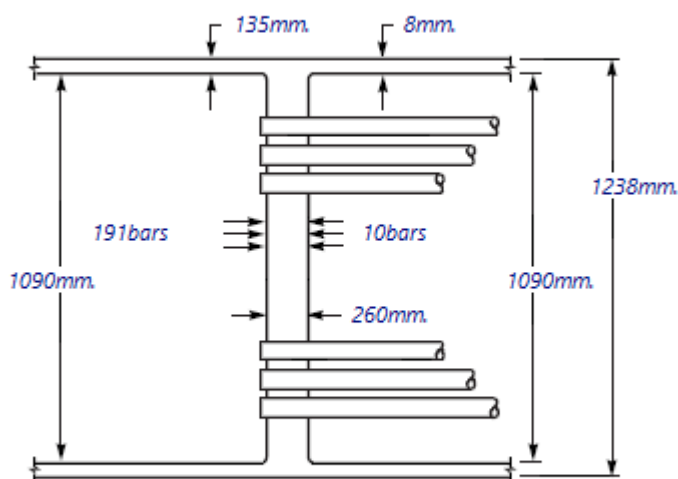


Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material	SA-965 F304	
Tubesheet Material UNS Number	S30400	
Tubesheet Allowable Stress at Temperature	S	129.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	260.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	1238.000 mm.
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	
Tubesheet Integral with	Both	
Tubesheet Extended as Flange	No	

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Material Group G

Elastic Mod. at Design Temperature	150.0 °C	0.18607E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TM-1 Material Group G

Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.



TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature 200.0 °C 0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature 21.1 °C 0.19512E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
= $0.5 \cdot 14.85 \cdot (\exp(191.0/(129.0 \cdot 1.0)) - 1) + 0$
= $1.185 + 0$
= 1.185 mm.

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{(Ro \cdot t)}$
= $4552.6 / \sqrt{(9.525 \cdot 2.1)}$
= 1017.929

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 / (19.05/2.1)$
= 0.0606

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t / Do)$
= $1.6 \cdot 0.061 \cdot 0.18235E+09(2.1/19.05)$
= 1950.121 N./mm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
= $1950.1/144.0$
= 13.542

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $144.0/442.5$
= 0.325

Ratio [AE]:

= Fhe / E
= $1950.1/0.18235E+09$
= 0.010694

Factor [m1]:

= $(\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys}))$
= $(\ln(0.325) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
= 0.244

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{psys}) / \ln(1 + \epsilon_{psys})^{m1}$
= $144.0(1 + 0.002) / \ln(1 + 0.002)^{0.244257}$
= 658.534 N./mm²

Curve Fitting Exponent for, Stainless Steel and Nickel Base Alloys [m2]:

= $m2f(m2f2 - R)$
= $0.75(1.0 - 0.325)$
= 0.506

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
= $(442.5 \cdot \exp(0.506)) / (0.506^{0.506})$
= 1035.962 N./mm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$
= $1.5 \cdot 0.3251.5 - 0.5 \cdot 0.3252.5 - 0.3253.5$



= 0.229

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	221	0.264	0.106E-04	-0.185E-04	0.211E-04	0.701E-04	0.119E+07	0.175E+00
2	110	-2.978	0.125E-05	0.123E-05	0.107E-04	-0.105E-04	0.307E+08	0.709E-02
3	165	-1.357	0.437E-05	0.262E-05	0.160E-04	-0.482E-05	0.536E+07	0.203E-01
4	138	-2.168	0.249E-05	0.230E-05	0.134E-04	-0.116E-04	0.141E+08	0.916E-03
5	152	-1.762	0.334E-05	0.274E-05	0.147E-04	-0.101E-04	0.897E+07	0.625E-02
6	145	-1.965	0.289E-05	0.255E-05	0.140E-04	-0.112E-04	0.114E+08	0.209E-02
7	141	-2.066	0.268E-05	0.243E-05	0.137E-04	-0.115E-04	0.127E+08	0.472E-03
8	140	-2.117	0.258E-05	0.236E-05	0.135E-04	-0.115E-04	0.134E+08	0.249E-03
9	140	-2.092	0.263E-05	0.240E-05	0.136E-04	-0.115E-04	0.130E+08	0.104E-03
10	140	-2.104	0.261E-05	0.238E-05	0.136E-04	-0.115E-04	0.132E+08	0.739E-04
11	140	-2.098	0.262E-05	0.239E-05	0.136E-04	-0.115E-04	0.131E+08	0.148E-04
12	140	-2.101	0.261E-05	0.238E-05	0.136E-04	-0.115E-04	0.132E+08	0.296E-04
13	140	-2.100	0.262E-05	0.239E-05	0.136E-04	-0.115E-04	0.132E+08	0.742E-05
14	140	-2.099	0.262E-05	0.239E-05	0.136E-04	-0.115E-04	0.131E+08	0.370E-05
15	140	-2.099	0.262E-05	0.239E-05	0.136E-04	-0.115E-04	0.132E+08	0.186E-05
16	140	-2.099	0.262E-05	0.239E-05	0.136E-04	-0.115E-04	0.131E+08	0.920E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$
 $= 2.407 - 0.741 \cdot Fic / F_y$
 $= 2.407 - 0.741 \cdot 140.62/144.0$
 $= 1.683$

Allowable Circumferential Compressive Stress [Fha]:

$= Fic/FS$
 $= 140.624/1.683$
 $= 83.537 \text{ N./mm}^2$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot Fha(t / Do)$
 $= 2 \cdot 835.319 (2.1/19.05)$
 $= 184.165 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1850 mm.
Allowable Internal Pressure at Corroded thickness	321.28 bars
Required Internal Design Pressure	191.00 bars
Allowable External Pressure at Corroded thickness	184.16 bars
Required External Design Pressure	10.00 bars
Required Thickness due to Shell Side pressure	0.3891 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 512.6 + 19.05$
 $= 1044.250 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19.05) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:



Ratio [ps]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1090.0/1044.2499$$

$$= 1.0438$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1090.0/1044.2499$$

$$= 1.0438$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/260.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t_t(E t_T / E) \cdot (St_T / S) \cdot (\rho); dt - 2t_t)$$

$$= \max(19.05 - 2 \cdot 2.1 (0.18235E+09 / 1.8235E+09) \cdot (129/129) \cdot (0); 19.05 - 2 \cdot 2.1)$$

$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 25.4 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 1044.25 \cdot 25.4) / (\pi \cdot 1044.25^2)) }$$

$$= 25.4000 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19.05) / 25.4$$

$$= 0.2500$$

$h/p = 10.236220$; $\mu^* = 0.250000$; $E^* = 39498860 \text{ KPa.}$

$E^*/E = 0.216611$; $\nu^* = 0.396457$

Note: As h/p (10.236) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{ (1090.0 + 8.0) 8.0 }$$

$$= 0.0194 \text{ 1/mm.}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.019 \cdot 1860724 \cdot 8.0^3 / (6(1 - 0.31^2))$$

$$= 3401415.2500 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1090.0 + 8.0) \cdot 3401415 / 260.0^3 \cdot (1 + 260.0 \cdot 0.019 + (260.0 \cdot 0.019)^2 / 2)$$

$$= 23850.9 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1090.0^2 / (4 \cdot 0.18607E+09 \cdot 8.0) \cdot (1 - 1090.0 \cdot 0.31 / (2(1090.0 + 8.0)))$$

$$= 0.16882E+00 \text{ mm./N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0438 \cdot 3401415 \cdot 0.0194 \cdot 0.168823 (1 + 260.0 \cdot 0.0194)$$

$$= 7004.1729 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ (D_c + t_c) t_c }$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{ (1090.0 + 135.0) 135.0 }$$

$$= 0.0045 \text{ 1/mm.}$$



Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.004 \cdot 1823496 \cdot 135.0^3 / (6(1 - 0.31^2))$$

$$= 3691688448.0000 \text{ bars} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1090.0 + 135.0 / \cos(0)) \cdot 0.36917E+1 / 260.0^3 \cdot (1.0 + 260.0 \cdot 0.004 \cdot (1.0 + 1.0) / 2 + (260.0 \cdot 0.004)^2 / 2 \cdot 1.0)$$

$$= 4374233.0 \text{ bars}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1090.0^2 / (4 \cdot 0.18235E+09 \cdot 135.0) \cdot (1 - 1090.0 \cdot 0.31 / (2(1090.0 + 135.0)))$$

$$= 0.10401E-01 \text{ mm} / \text{N} / \text{mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.044 \cdot 0.36917E+1 [0.010401(1.0 + 0.004 \cdot 1.0 \cdot 260.0) - 0(1.0 / 260.0 + 1.0 \cdot 0.004 / 2)]$$

$$= 38641.2227 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1238.0 / 1044.2499$$

$$= 1.1855$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 39498860 \cdot (23850.92 + 4374233 + 0.18235E+09 \cdot \ln(1.19))$$

$$= 7.1945$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 1044.25^2 / 16 \cdot [(1.044 - 1) \cdot (1.044^2 + 1) \cdot 10.0 - (1.044 - 1) \cdot (1.044^2 + 1) \cdot 191.0]$$

$$= -1129292.0000 \text{ bars} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps$$

$$= -1129292.0 + 38641.223 \cdot 191.0 - 7004.173 \cdot 10.0$$

$$= 6181140.0000 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F(Ps - Pt) / (1 + F)$$

$$= (6181140) - 1044.25^2 / 32 \cdot 7.195(10.0 - 191.0) / (1 + 7.19)$$

$$= 6169518.5000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 6169518 + 1044.25^2 / 64 \cdot (3 + 0.396) \cdot (10.0 - 191.0)$$

$$= -4304989.5000 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:



$$\begin{aligned}
 &= \max(|M_p|; |M_o|) \\
 &= \max(|6169518|; |-4304989.5|) \\
 &= 6169518.5000 \text{ bars} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 6169518 / (0.25 \cdot (260.0 - 6.0)^2) \\
 &= 229.5199 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 129.0 \\
 &= 258.00 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 6144937 / (0.25 \cdot (245.2289 - 6.0)^2) \\
 &= 257.7072 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 245.2289 + 0 + 0 \\
 &= 245.2289 \text{ mm.}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(245.2289; 221.2278) \\
 &= 245.2289 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |10.0 - 191.0| \\
 &= 181.000 \text{ bars}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.25 \cdot 260.0 / 1044.25 \cdot \min(0.8 \cdot 129.001; 0.533 \cdot 144.0) \\
 &= 19.110 \text{ N./mm}^2
 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\
 &= 1 / (4 \cdot 0.25 \cdot 76.752) \cdot (4 \cdot 7543.9 / 3216.83) \cdot |10.0 - 191.0| + 0 \\
 &= 221.2278 \text{ mm.}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 8.000mm.

Min. Shell len. adjacent to the tubesheet is: 168.086 mm.

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$\begin{aligned}
 &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\
 &= 1090.0^2 / (4 \cdot 8.0 (1090.0 + 8.0)) \cdot 10.0 \\
 &= 33.8163 \text{ N./mm}^2
 \end{aligned}$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$\begin{aligned}
 &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot \\
 &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\
 &= 6 \cdot 76468.61 / 8.0^2 \cdot [0.019 \cdot 0.1686 + 6(1 - 0.396) / 39498860 \cdot \\
 &\quad 1044.25 / 260.0^3 \cdot (1 + 260.0 \cdot 0.019 / 2) \cdot (6169518 + 1044.25^2 / 32 \cdot \\
 &\quad (10.0 - 191.0))] \\
 &= 104.353 \text{ N./mm}^2
 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:



$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s \text{)} \\ = 138.169 \text{ N./mm}^2 \text{ (Should be } \leq 207.00 \text{)}$$

Note: For a given Channel thickness of: 135.000 mm.
Min. Channel len. adjacent to the tubesheet is: 690.482mm.

Axial Channel Membrane Stress $[\sigma_{c,m}]$:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t \\ = 1090.0^2 / (4 \cdot 135.0 (1090.0 + 135.0) \cdot \cos(0)) \cdot 191.0 \\ = 34.307 \text{ N./mm}^2$$

Axial Channel Bending Stress $[\sigma_{c,b}]$:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - \\ 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + \\ h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ = 6 \cdot 82994360 / 135.0^2 \cdot [(0.010401 \cdot 1.0 \cdot 0.004 - 0 \cdot 1.0) \cdot 191.0 - \\ 6((1 - 0.396) / 39498860) \cdot 1044.25 / 260.0^3 \cdot (1.0 + \\ 0.004 \cdot 260.0 / 2 \cdot 1.0) \cdot (6169518 + 1044.25^2 / 32 \cdot (10.0 - 191.0))] \\ = 107.592 \text{ N./mm}^2$$

Channel Membrane + Bending Stress $[\sigma_c]$:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c \text{)} \\ = 141.899 \text{ N./mm}^2 \text{ (Should be } \leq 193.50 \text{)}$$

Tube Metal Area $[A_t]$:

$$= \pi (d_o - t) t \\ = \pi (19.05 - 2.1) 2.1 \\ = 1.118 \text{ cm}^2$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength $[F_t]$:

$$= \pi \cdot t (d_o - t) S \\ = \pi \cdot 2.1 (19.05 - 2.1) 129.001 \\ = 1470.970 \text{ Kgf}$$

Groove Weld Strength $[F_g]$:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ = 0.85 \cdot \pi \cdot 2.1 (19.05 + 0.67 \cdot 2.1) 129.0 \\ = 1470.9697 \text{ Kgf}$$

Weld Strength Factor $[f_w]$:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ = 129.001 / 129.001 \\ = 1.000$$

Design Strength Ratio $[f_d]$:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength $[F_f]$:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)} \\ = 0.55 \cdot \pi \cdot 2.1 (19.05 + 0.67 \cdot 2.1) 129.0 \\ = 976.4246 \text{ Kgf}$$

Minimum Length of Weld $[a_r]$:

$$= 2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o} \\ = 2 \cdot \sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.1 \cdot (19.05 - 2.1) 1.0 \cdot 1.0) - 0.75 \cdot 19.05} \\ = 2.552 \text{ mm.}$$

Minimum Required Fillet Weld Leg $[a_{fr}]$: 1.2759 mm.
Minimum Required Groove Weld Leg $[a_{gr}]$: 1.2759 mm.

Tube-Tubesheet Jt allowable, 1470.97 is $\geq$ tube strength 1470.97 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
--------------------	--------	-----------	-----------



Tubesheet bend. stress	229.5	<=	258.0	N./mm ²	Ok
Tubesheet shear stress	65.3	<=	76.8	N./mm ²	Ok
Stress in Shell at Tubesheet	138.2	<=	207.0	N./mm ²	Ok
Stress in Channel at Tubesheet	141.9	<=	193.5	N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	245.229	260.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.276	2.100	Ok
Tube-Tubesheet Groove Weld Leg :	1.276	2.100	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	251.123 ...	241	258	69 77	Ps+Pt D1	Ok
D2uc	62.181 ...	11	258	4 77	Ps+Pt D2	Ok
D3uc	245.229 ...	230	258	65 77	Ps+Pt D3	Ok
Max:	251.1227 ... mm.	0.933	0.898	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		2	207	146	194	Ok
D2uc		141	207	4	194	Ok
D3uc		138	207	142	194	Ok
Max		0.679	0.755			

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	251.123	260.000	Ok
Tubesheet thickness flanged extension :	0.000	64.000	Ok
Tube thickness :	1.185	2.100	Ok
Tube-tubesheet fillet weld leg :	1.276	2.100	Ok
Tube-tubesheet groove weld leg :	1.276	2.100	Ok

Given Shell Thickness: 8.0000 mm.
 Given Channel Thickness: 135.0000 mm.

Min Shell length of thk, (8.000) adj. to tubesheet: 168.086 mm.

Min Channel length of thk, (135.000) adj. to tubesheet: 690.482 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	204.742	1.000	231.080	1.000
Tubesheet Shear Stress	212.721	1.000	212.721	1.000
Shell Stress (Axial, Junction)	212.721	0.013	29.429	0.999
Tubesheet-Channel Junction	212.721	0.841	29.429	0.127
Tube Pressure Stress	321.280	1.000	184.164	1.000
Minimum MAWP	204.742		29.429	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	219.020	1.000	247.198	1.000
Tubesheet Shear Stress	302.826	1.000	302.826	1.000
Shell Stress (Axial, Junction)	302.826	1.383	29.429	0.999
Tubesheet-Channel Junction	302.826	1.383	29.429	0.119
Tube Pressure Stress	343.686	1.000	251.548	1.000
Minimum MAPnc	219.020		29.429	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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